

Lexicographic orders of the well-orderings of the reals

Ordinal spectra, coding length and representation rank,
finite-tail geometry, characters, gaps, and self-absorption

Research report merged from three independent manuscripts
prepared for Vladimir Reshetnikov (sources 02, 06 and 07 of batch 76)

Sources dated 2 October 2026; merged 2 October 2026

Abstract

Let $\kappa = 2^{\aleph_0}$, regarded as an initial ordinal. A well-ordering of $\mathbb{R}$ has a unique exhaustive enumeration by an ordinal of cardinality κ . We compare two such enumerations at their first unequal real entry. The resulting linear order $\mathcal{W}$ has cardinality 2^κ , contains every linear order of cardinality at most κ , and contains an ordinal β exactly when $\beta < \kappa^+$. Nevertheless, its least binary lexicographic coding length is κ^+ : it contains every binary lexicographic cube of smaller length but does not contain the cube of length κ^+ .

The restriction $\mathcal{W}_\kappa$ to enumerations of length exactly κ is quite different. It is equimorphic with $\{0, 1\}^\kappa$ in lexicographic order, as is the actual ultrafilter-enumeration index in the Kanovei–Shelah construction. We prove the exact fixed-type classification $\mathcal{W}_{\lambda+n} \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$ for limit λ of cardinality κ , characterize every adjacent pair in $\mathcal{W}$, compute several order-topological invariants, and prove the sharp absorption theorem $\mathcal{W}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}$ if and only if $\gamma < \kappa^+$. We also analyze Dedekind completion and formulate further embedding, definability, and formalization problems.

[merge] This report is built from three manuscripts written independently on the same question and delivered together in batch 76: source 02 (the base, whose abstract is the two paragraphs above), source 06 and source 07. They prove the same core three times. Each shared theorem is printed once, with the other sources' arguments marked as second or third routes where the argument differs. The present stage prints source 02 in full, with those routes and the theorem crosswalk of [section C](#); the results proved only by sources 06 and 07 are listed in the crosswalk with their source numbers.

Conventions and status. All main theorems are in ZFC. All embeddings are order embeddings, not necessarily continuous maps. Equimorphism means embeddability in both directions, not isomorphism. Proofs are supplied below; they have not been independently refereed or checked by a proof assistant. The results are presented as a research synthesis with explicit derivations, not as a claim of established historical priority.

[merge] **Status of the merged report.** AI-assisted (all three sources are “prepared for Vladimir Reshetnikov”; none carries AI wording in its author line), unrefereed, and not formalized. None of the three sources claims historical priority, peer review, or machine verification; their finite checks are illustrations of finite instances, not evidence for the transfinite assertions.

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1 About this report

1.1 Three independent manuscripts

[merge] This section, the notation table, the crosswalk of [section C](#) and every paragraph tagged [merge] were added when the three manuscripts of [table 1](#) were merged. Everything untagged in [sections 2 to 16](#) and [sections A and B](#) is source 02 as delivered, with its labels prefixed by `lwo:`. A reference such as “06:171” means line 171 of the main `.tex` file of source 06 as delivered (archive in the arrival commit 6914ccca6; the manuscripts of sources 06 and 07 are not shipped with this report).

Source	Archive (batch 76)	Delivered title	Size
02 (base)	lexicographic_well_orderings_of_reals.zip	<i>Lexicographic Orders of Well-Orderings of the Continuum: ordinal spectra, lexicographic complexity, finite-tail geometry, and self-absorption</i>	1,826 lines; 24 pp.
06	Lexicographic_Well_Orderings_of_Reals (1).zip	<i>The Lexicographic Order of All Well-Orderings of the Real Line: universality, exact ordinal spectra, finite tails, local characters, and Dedekind gaps</i>	1,522 lines; 23 pp.
07	lexicographic_well_orderings_research.zip	<i>Lexicographic Orders of Real Well-Orderings: ordinal universality, representation rank, products, and cut geometry</i>	1,521 lines; 21 pp.

Table 1: [merge] The three sources. The number of a source is its batch-76 manuscript number. All three are dated 2 October 2026, are “prepared for Vladimir Reshetnikov”, and pin the ProveIt revision 6ea60e3677bd847a00baf023c3c532c83884530e (source 02: [section B](#); source 06: its Section 1.1 and audit; source 07: its Section 13.2 and audit). They arrived in commit 6914ccca6 and were placed in commit 2a04b60f2.

The three texts are *independent*: they share no member file, their word 8-gram overlap is 0.65–0.75 % pairwise, and their titles, notation and proofs were written separately. None is a version of another, and none supersedes another. They nevertheless prove the same core—prefix freeness, cardinality 2^κ , universality for orders of size at most κ , the exact ordinal spectrum below κ^+ , the equimorphism $\mathcal{W}_\kappa \simeq_e \mathbb{B}_\kappa$, the adjacency criterion, the finite blocks of size $n!$, and the topology of $\mathcal{W}$ and of $\mathcal{W}_\kappa$ —three times. Sources 02 and 07 further share the coding rank κ^+ , the real powers, power absorption, the fixed-length square and the Kanovei–Shelah index; sources 06 and 07 share the point-character spectrum, the gap spectrum of $\mathcal{W}$ and the dense quotient. Source 02 is the base because it has the widest scope: every stratum $\mathcal{W}_\alpha$, the Kanovei–Shelah index $\mathcal{A}$, general alphabets in its coding-length theorem, and completion invariants.

1.2 How the merge is printed

[merge] Each theorem proved by more than one source is stated once. Where another source proves it by the same argument, a short note after the proof says so (“also source 06, Theorem 5.2, 06:465, same argument”). Where the argument differs, the other proof is printed as a marked *second route* (or *third route*), credited to its source. No result, proof, example, remark, question or limitation of any source is dropped: the crosswalk ([table 4](#)) lists every numbered statement of the three sources and where it stands.

Two cross-repository duplications are also recorded at their points of use. Source 02’s strict cube hierarchy ([theorem 5.3](#)) re-proves Corollary `cor:strict` of the ProveIt report *Every ordinal as a point-separating game value*; that corollary is cited first and source 02’s cylinder proof is a second route. Source 07’s strictness of real powers ([corollary 5.6](#)) is the ordinal case of a classical theorem of Kuhlmann [6, Corollary 2.4], as source 07 itself says.

1.3 Questions one source answers for another

[merge] Each source left open a question that another source answers. These answers are printed as theorems, crediting the answering sources, and the questions are re-scoped in [section 16](#).

- (i) Source 06 proves that $\mathcal{W}$ does not embed into $\mathcal{W}_\kappa$ only under $2^{<\kappa} < 2^\kappa$ (06:1053, its Corollary 9.5) and asks for an unconditional comparison (06:1332, its Research question 14.1). Sources 02 and 07 prove it in ZFC: [corollary 8.3](#) (source 02) and source 07's Corollary 5.4 (07:525), which also excludes every fixed real power $\mathbb{R}_{\text{lex}}^\gamma$ with $\gamma < \kappa^+$.
- (ii) Source 02 asks for the exact spectra of cuts and point characters (02:1635–1646). Sources 06 and 07 compute the point-character spectrum and the gap spectrum of $\mathcal{W}$ (06:913, 06:802; 07:861, 07:941), and source 07 the gap spectrum of $\mathcal{W}_\kappa$ (07:996).
- (iii) Source 02 asks whether the finite-condensation quotient $\mathcal{W}/\sim_{\text{fin}}$ is equimorphic with $\mathcal{W}$ (02:1604–1607). Sources 06 and 07 prove that it is (06:1133, 07:1072).

1.4 Notation

[merge] The three sources use different letters for the same objects and the same letters for different objects. The merged text uses the symbols of [table 2](#); every symbol taken from source 06 or 07 and renamed here is listed in its last column. No normalization changes: every renaming is a change of letter only.

Table 2: [merge] Notions and symbols. “False reading” marks a tempting confusion.

Notion	Here	Sources, and readings to avoid
Continuum as an initial ordinal	$\kappa = \mathfrak{c} = 2^{\aleph_0}$	02 and 06: κ (06 also writes $\mathfrak{c}$); 07: κ (macro $\backslash\kappa$). False reading: κ^+ is the successor <i>cardinal</i> , not the ordinal $\kappa + 1$.
All well-orderings of $\mathbb{R}$	$\mathcal{W} = \mathcal{W}(\mathbb{R})$	02, 06: $\mathcal{W}$; 07: $\mathcal{W}$ (macro $\backslash\mathcal{W}0$). The order compares enumerations at the first unequal entry in the <i>usual</i> order of $\mathbb{R}$; it is not an order of ordinal types.
Enumerations of an ordered set X	$\mathcal{W}(X), \mathcal{W}_\alpha(X)$	02: $ X = \lambda$; 06, 07: $ X = \nu$. Kept as printed in each source's text.
Stratum of length α	$\mathcal{W}_\alpha$	02: $\mathcal{W}_\alpha$; 06: $\mathcal{W}_\beta$.
Minimal-type slice	$\mathcal{W}_\kappa$	06: $\mathcal{P}(\backslash\text{PP})$; 07: $\mathcal{P}(\backslash\text{Perm})$. False reading: source 07's $\mathcal{P}$ is <i>not</i> the power set $\mathcal{P}(\mathbb{N})$, which source 07 also uses.
Kanovei–Shelah index	$\mathcal{A}$, fibres $\mathcal{A}_U$	02: $\mathcal{A}$; 07: $\mathcal{A}(\backslash\text{KS})$; ultrafilter: 02 U , 07 D , here U .
Binary cube	$\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$	06: $({}^\theta 2, <_{\text{lex}})$; 07: 2_{lex}^θ . False reading: 2^κ without a subscript is a cardinal.
Real power	$\mathbb{R}_{\text{lex}}^\gamma$	06: $({}^\gamma \mathbb{R}, <_{\text{lex}})$; 07: $\mathbb{R}_{\text{lex}}^\gamma$. All functions, no finite-support restriction.
Binary coding length	$\ell_2(L)$	02 only: $\min\{\theta : L \hookrightarrow \mathbb{B}_\theta\}$. It equals the game value $\text{ps}(L^{\text{dd}})$ of the point-separating report (its Theorem thm:embedding).
Real representation rank	$\text{repr}_{\mathbb{R}}(L)$	07 only, after Giarlotta: $\min\{\alpha : L \hookrightarrow \mathbb{R}_{\text{lex}}^\alpha\}$. Both invariants are kept: $\text{repr}_{\mathbb{R}}(L) \leq \ell_2(L) \leq \omega \cdot \text{repr}_{\mathbb{R}}(L)$; they agree on $\mathcal{W}(\kappa^+)$ and on $\mathcal{W}_\kappa$ and $\mathcal{A}(\kappa)$, but not in general ($\text{repr}_{\mathbb{R}}(\mathbb{R}) = 1$, $\ell_2(\mathbb{R}) = \omega$).

Notion	Here	Sources, and readings to avoid
Equimorphism	$L \simeq_e M$	02: $\simeq_e$; 07: $\equiv_{\text{emb}}$; 06: “mutually order-embeddable”. False reading: not isomorphism.
Order embedding	$L \hookrightarrow M$	06: $\hookrightarrow_{\text{ord}}$. Not continuous and not convex unless said.
Reverse order	L^{op}	06: L^* ; 07: L^{op} .
Prefix cylinder	C_s, C_p	02: C_s ; 06: cone $C(p)$; 07: $[p]$.
Residual set after a prefix p	$Y_p = X \setminus \text{ran } p$	02: Y ; 06: A_p ; 07: $S \setminus \text{ran } p$. False reading: source 06’s A_p is unrelated to the index $\mathcal{A}$.
First disagreement	$\delta, \Delta(e, f)$	02, 07: δ ; 06: $\Delta(e, f)$.
Lower/upper branch positions	$D^-(e), D^+(e)$	06 and 07, same meaning. False reading: unrelated to source 02’s $D_{\kappa, m} = \mathbb{B}_{\kappa} \times_{\text{lex}} \{0, \dots, m-1\}$.
One-sided point characters	$\chi^-(e), \chi^+(e)$	06: $\chi^{\mp} (\backslash \text{cfm}, \backslash \text{cfp})$; 07: $\chi^{\mp} (\backslash \text{chm}, \backslash \text{chp})$; 02: “left and right local cofinalities”.
cf κ	cf κ	07: θ . Renamed because θ is an ordinal variable here ($\mathbb{B}_{\theta}$).
$2^{<\kappa} = \sup_{\lambda < \kappa} 2^{\lambda}$	$2^{<\kappa}$	07: τ .
Density, cellularity, weight	d, cell, w	02: d, cell; 06, 07: d, c, w . Cellularity is written cell, never c , to avoid $\mathfrak{c}$.
Infinite regular cardinals $\leq \kappa$	$\text{Reg}(\leq \kappa)$	06: “infinite regular $\rho \leq \kappa$ ”; 07: $\text{Reg}_{\infty}(\leq \kappa)$.
Gap and its character	(L, U) , (cf L , ci U); GapChar	06: (L, U) ; 07: (A, B) ; 02 (proposition 14.1): (L, R) . A gap has no endpoint on either side; a jump is not a gap.
Finite-condensation quotient	$\mathcal{W}/\sim_{\text{fin}}$	02: $\mathcal{W}/\sim_{\text{fin}}$; 06: D ; 07: $\mathcal{W}_{\text{fc}}$. False reading: source 06’s D is not $D^{\pm}(e)$.
Injections $\kappa \rightarrow \mathbb{R}$	$\mathcal{I}$	06: J (not necessarily onto).
Well-ordering-based ultrafilter index	$\mathcal{J}$	07: J (bijections $\kappa \rightarrow \mathcal{P}(\mathbb{N})$). False reading: sources 06 and 07 both write J for these two different orders.
Bounded surreal sign order	$S_{<\theta}$	06 only.

1.5 Non-claims of the three sources

[merge] Every limitation stated by a source is kept where the source states it; together they are:

- No source claims historical priority (02: section B; 06: its abstract and Section 1.1; 07: its abstract and Section 13.2). Source 07 identifies its real-power strictness as Kuhlmann’s classical theorem; source 06 attributes the characteristic-function coding to the standard literature (Moore’s Section 2); source 02 cites Knoblauch for the terminology of intrinsic length.
- The proofs are written ZFC arguments. None has been refereed or checked in Lean or Rocq. The finite checks of all three sources test finite instances only (section A).
- Repository and literature searches were targeted, not exhaustive (02: section B; 06 and 07: their shipped audits).
- The research questions are proposed continuations, not an authenticated list of published open problems.
- Source 06 does not classify all gaps of $\mathcal{W}_{\kappa}$ through its missing-branch construction, does not describe the automorphism group, and does not claim that full choice is necessary (its audit, Section 5).

- No source gives an intrinsic criterion for an arbitrary linear order to embed into $\mathcal{W}$; absence of κ^+ and its reverse is necessary and is not claimed sufficient (02, 06 and 07 all say so).

Placement of this report in the Cardinals collection, beside the Lean development of the Cardinals project, confers no formal status: no statement of this report has a Lean or Rocq counterpart.

2 Three orders that must be distinguished

2.1 The literal order of all well-orderings

Throughout the real case, put

$$\kappa = |\mathbb{R}| = 2^{\aleph_0},$$

and identify this cardinal with its initial ordinal. For an ordinal α of cardinality κ , let

$$\mathcal{W}_\alpha = \{e : \alpha \longrightarrow \mathbb{R} : e \text{ is a bijection}\}, \quad \mathcal{W} = \bigcup_{\kappa \leq \alpha < \kappa^+} \mathcal{W}_\alpha.$$

A well-order relation on $\mathbb{R}$ determines exactly one member of $\mathcal{W}$: its increasing enumeration in that well-order. Conversely, $e \in \mathcal{W}_\alpha$ defines the relation

$$r \prec_e s \iff e^{-1}(r) < e^{-1}(s).$$

Thus $\mathcal{W}$ really represents all well-orderings, not merely their ordinal isomorphism types. The order in which we compare their entries is the *usual* order of the real numbers, not one of the well-orders being compared.

For distinct $e, f \in \mathcal{W}$, write

$$e <_{\text{lex}} f \iff e(\delta) < f(\delta), \quad \delta = \min\{\xi < \min(\text{dom } e, \text{dom } f) : e(\xi) \neq f(\xi)\}. \quad (1)$$

The existence of δ is proved in [lemma 4.1](#). In particular, there is no choice of an “end-of-word” convention hidden in this definition.

One could instead sort by ordinal length first. That would be a different order: choosing one enumeration of each possible length would then give an increasing sequence of type κ^+ . This is impossible for (1). Another different construction is to compare characteristic functions of well-order relations using a chosen well-order of $\mathbb{R}^2$. Neither alternative is used here.

2.2 The fixed-length restriction

The suborder $\mathcal{W}_\kappa$ contains only those well-orderings whose ordinal type is the initial ordinal κ . It is a substantial suborder, but it is not all of $\mathcal{W}$. For example, types $\kappa + 1$, $\kappa + 2$, and $\kappa \cdot 2$ also occur in $\mathcal{W}$.

This distinction cannot be dismissed on cardinality grounds. Both $\mathcal{W}_\kappa$ and $\mathcal{W}$ have cardinality 2^κ , yet $\mathcal{W}$ does not embed into $\mathcal{W}_\kappa$. Indeed, no union of strata with a bounded set of ordinal lengths can contain an embedded copy of $\mathcal{W}$.

2.3 The index in the checked Kanovei–Shelah paper

Kanovei and Shelah’s 2004 paper constructs a definable, countably saturated elementary extension of the reals. Its Section 1 uses the index

$$\mathcal{A} = \{a : \kappa \longrightarrow \mathcal{P}(\mathbb{N}) : \text{ran } a \text{ is an ultrafilter on } \mathbb{N}\}, \quad (2)$$

ordered lexicographically; subsets of $\mathbb{N}$ are themselves compared by their binary characteristic sequences [1, Section 1]. These maps need not be injective. The definition includes principal ultrafilters. For an ultrafilter U , write $\mathcal{A}_U = \{a \in \mathcal{A} : \text{ran } a = U\}$.

Thus the checked source uses fixed-length ultrafilter enumerations, not literally $\mathcal{W}$. The two constructions share the principle of ordering *presentations* without first selecting a presentation. We study both rather than identifying them.

2.4 Connection to ProveIt and to existing terminology

The ProveIt report *Every ordinal as a point-separating game value* contains a binary-cylinder argument and an order-embedding interpretation of lexicographic coding [4]. We use that line of thought, but give the needed cube argument independently in section 5. None of that report's claims about compact T_1 spaces is needed here.

Minimum ordinal lengths of binary lexicographic representations are also studied under the name *intrinsic length* in work of Knoblauch [2]. We fix our precise convention in definition 5.4; no classification theorem from that literature is assumed. The targeted source search did not establish a prior full classification of the specific order $\mathcal{W}$ considered here. This is a limitation of the bibliography, not a claim that such a classification is absent from the literature.

3 Main conclusions at a glance

Let $\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$, for every ordinal θ . An expression such as $\mathbb{R}_{\text{lex}}^\gamma$ always has ordinal-indexed coordinates, read from left to right. We use $L \simeq_e M$ for equimorphism.

Theorem 3.1 (Main embedding spectrum). *The following statements hold.*

- (i) $|\mathcal{W}| = |\mathcal{W}_\kappa| = |\mathcal{A}| = 2^\kappa$. Every linear order of cardinality at most κ embeds into each of these orders.
- (ii) For every ordinal β and each $L \in \{\mathcal{W}, \mathcal{W}_\kappa, \mathcal{A}\}$,

$$\beta \hookrightarrow L \iff \beta < \kappa^+.$$

The same equivalence holds for reverse ordinals.

- (iii) $\mathcal{W}_\kappa \simeq_e \mathcal{A} \simeq_e \mathcal{A}_U \simeq_e \mathbb{B}_\kappa$ for every ultrafilter U on $\mathbb{N}$.
- (iv) $\mathbb{B}_\theta \hookrightarrow \mathcal{W}$ exactly when $\theta < \kappa^+$, whereas $\mathbb{B}_\theta \hookrightarrow \mathcal{W}_\kappa$ exactly when $\theta \leq \kappa$. The latter criterion also holds with $\mathcal{A}$ in place of $\mathcal{W}_\kappa$.
- (v) $\mathbb{R}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}$ exactly when $\gamma < \kappa^+$. It embeds into $\mathcal{W}_\kappa$ or $\mathcal{A}$ exactly when $\gamma \leq \kappa$.
- (vi) $\mathcal{W}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}$ exactly when $\gamma < \kappa^+$. For $0 < \gamma < \kappa^+$ the two orders are equimorphic. In contrast, $\mathcal{W}_\kappa \times_{\text{lex}} \mathcal{W}_\kappa \not\hookrightarrow \mathcal{W}_\kappa$, and $\mathcal{A} \times_{\text{lex}} \mathcal{A} \not\hookrightarrow \mathcal{A}$.

These assertions are proved in sections 6 to 8 and 10. [merge] Items (i)–(iii) are also the opening theorems of sources 06 (Theorem 1.1, 06:102) and 07 (Theorem 2.1, 07:187); items (iv)–(vi) are also in source 07's Theorem 2.1, in real-power form. The exact refinement for every fixed type, proved in section 9, is

$$\mathcal{W}_{\lambda+n} \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} \{0, \dots, n! - 1\}, \quad |\lambda| = \kappa, \quad \lambda \text{ limit}, \quad n < \omega.$$

Moreover, two strata compare by their limit parts first and by their final factorial block sizes second; see corollary 9.4. The structural distinction is summarized in table 3. Here $2^{<\kappa} = \sup_{\lambda < \kappa} 2^\lambda$, with cardinal

Invariant	$\mathcal{W}_\kappa$	$\mathcal{A}$	$\mathcal{W}$
Cardinality	2^κ	2^κ	2^κ
Ordinals that embed	$\beta < \kappa^+$	$\beta < \kappa^+$	$\beta < \kappa^+$
Binary coding length ℓ_2	κ	κ	κ^+
Dense linear order	yes	yes	no
Least or greatest element	neither	neither	neither
Cofinality	ω	$\text{cf}(\kappa)$	ω
Cointinality	ω	ω	ω
Density and cellularity	$2^{<\kappa}$	$2^{<\kappa}$	2^κ
Isolated points	none	none	2^κ
Self-dual	yes	no	yes
Unfilled (ω, ω) cut	yes	yes	yes
Absorbs its lexicographic square	no	no	yes

Table 3: Equal ordinal spectra do not imply equal lexicographic complexity. The assertions about $\mathcal{A}$ refer to the full index (2).

exponents; cell is the supremum of the sizes of pairwise disjoint families of nonempty open sets. All topologies in the table are order topologies.

Under CH the ordinal cutoff is ω_2 : every ordinal below ω_2 embeds, including $\omega_1 \cdot 2$ and ω_1^2 , but ω_2 does not. Without CH, the answer is still exactly κ^+ , not an absolute cutoff at ω_2 . For example, if $\mathfrak{c} = \aleph_2$, then ω_2 embeds and ω_3 does not. These are conditional substitutions into the ZFC theorem, not additional assumptions in its proof.

4 Exhaustive enumeration orders

Several central facts hold for any infinite linearly ordered set X , not just the real line. Write $\lambda = |X|$ and let $\mathcal{W}(X)$ consist of all bijections from ordinals of cardinality λ onto X , with the same first-difference comparison.

Lemma 4.1 (Prefix-freeness). *No member of $\mathcal{W}(X)$ is a proper initial segment of another. Consequently two distinct members have a first disagreement in the common part of their domains, and first-difference comparison is a linear order.*

Proof. Suppose $e : \alpha \rightarrow X$ is an initial segment of the injective map $f : \beta \rightarrow X$, where $\alpha < \beta$. Since e is onto, the value $f(\alpha)$ already occurs at some coordinate below α , contrary to injectivity. If two enumerations agree throughout their common domain, one is therefore equal to the other.

For transitivity, suppose $e < f < g$, with first disagreements δ and ε . Before $\min(\delta, \varepsilon)$ all three agree. If the coordinates are unequal, the earlier comparison gives $e < g$ there. If they are equal, $e(\delta) < f(\delta) < g(\delta)$ gives the same conclusion. Trichotomy follows from the linear order on X . $\square$

[merge] Also source 06, Lemma 2.1 and Proposition 2.3 (06:171, 06:198), and source 07, Section 1.2 (07:132–141), by the same argument.

Definition 4.2. For an injective ordinal sequence s of elements of X , its cylinder is

$$C_s = \{e \in \mathcal{W}(X) : s \subseteq e\}.$$

We consider only prefixes that have an exhaustive extension. An exhaustive prefix has a singleton cylinder.

Lemma 4.3 (Residual-set geometry). *Every cylinder is convex. If s is not exhaustive and $Y = X \setminus \text{ran } s$, then concatenation gives an order isomorphism*

$$\mathcal{W}(Y) \longrightarrow C_s, \quad t \longmapsto s \smallfrown t.$$

For finite Y of size m , the cylinder has exactly $m!$ points.

Proof. A point between two extensions of s cannot first disagree with s before its end: that disagreement would put it on the same strict side of both extensions. The remaining assertions follow because every exhaustive extension must enumerate precisely Y after s , and first differences in the tail are unchanged by concatenation. $\square$

[merge] Sources 06 (Lemma 3.2, 06:246) and 07 (Lemma 3.1, 07:259) prove the same lemma and add its next-value decomposition: if $Y_s = X \setminus \text{ran } s$ is nonempty, then

$$C_s \cong \sum_{a \in Y_s}^{<X} \mathcal{W}(Y_s \setminus \{a\}), \quad (3)$$

a lexicographic sum over Y_s in its given order, whose summand indexed by a is the child cylinder $C_{s \smallfrown \langle a \rangle}$, with $\mathcal{W}(\emptyset)$ a one-point order. Sorting the members of C_s by their next entry proves it. Source 06 also observes that concatenation uses ordinal addition, so no ordinal subtraction convention is needed (06:242–244).

In particular, a long prefix may have already used all but two or three reals. This is why the density of the usual real order does *not* imply density of $\mathcal{W}$.

Proposition 4.4 (Cardinality). *For every infinite linearly ordered set X of cardinality λ ,*

$$|\mathcal{W}(X)| = |\mathcal{W}_\lambda(X)| = 2^\lambda.$$

Proof. A well-order relation is a subset of X^2 , giving the upper bound $2^{|X^2|} = 2^\lambda$. For the lower bound, partition X into λ two-element sets $\{a_\xi, b_\xi\}$, with $a_\xi < b_\xi$. For $x \in \{0, 1\}^\lambda$, enumerate the pair at coordinate ξ as (a_ξ, b_ξ) if $x(\xi) = 0$, and as (b_ξ, a_ξ) otherwise. The concatenation has length $2 \cdot \lambda = \lambda$ and enumerates all of X . Different bit sequences give different enumerations. In fact, this is an order embedding $\mathbb{B}_\lambda \hookrightarrow \mathcal{W}_\lambda(X)$. $\square$

[merge] Also source 06, Lemma 4.1 and Corollary 4.2 (06:331, 06:363), and source 07, Lemma 4.1 and Corollary 4.6 (07:329, 07:417), by the same pair-swap code. Source 06 notes that the multiplication in $2 \cdot \nu = \nu$ is ordinal multiplication, the two positions of a block being the less significant coordinates (06:358–361). Two definitions that are *not* used are ruled out by source 06 (06:220–230): comparing characteristic functions of relations at the “least real where they differ” fails, since a set of real disagreement coordinates need not have a least member; and the order does not descend to isomorphism types, since two enumerations of any two admissible types can be made to compare either way by their first values.

5 Binary cubes and the cylinder obstruction

Lemma 5.1 (Cut coding). *Every linear order L of cardinality at most an infinite cardinal λ embeds into $\mathbb{B}_\lambda$.*

Proof. Choose a well-ordered list $(t_\xi)_{\xi < \mu}$ of the elements of L , where $\mu \leq \lambda$. For $x \in L$, put

$$c_x(\xi) = \begin{cases} 1 & t_\xi \leq_L x, \\ 0 & t_\xi >_L x, \end{cases} \quad (\xi < \mu),$$

and put $c_x(\xi) = 0$ for $\mu \leq \xi < \lambda$. If $x < y$, then $c_x \leq c_y$ coordinatewise, and the coordinate listing y distinguishes them. Their first unequal bits are consequently 0 and 1. $\square$

[merge] Also source 06, Lemma 4.3 (06:373), and source 07, Lemma 4.2 (07:339), coding x by the characteristic function of $\{t : t < x\}$ instead of $\{t : t \leq x\}$; the argument is the same. Source 06 attributes this coding to the standard literature (Moore’s representation of Aronszajn lines, [5, Section 2]) and stresses that it uses a well-ordering of the coordinate set, not one compatible with L (06:385–387).

Lemma 5.2 (A cut preserves a full child). *Let C_s be a full prefix cylinder in $\mathbb{B}_\theta$, with $\text{dom } s < \theta$. If D is an initial segment of C_s , then either $C_{s \smallfrown 0} \subseteq D$ or $C_{s \smallfrown 1} \subseteq C_s \setminus D$. The same assertion holds for cylinders $C_s \times F$ in $\mathbb{B}_\theta \times_{\text{lex}} F$, where F is any nonempty linear order.*

Proof. If the first inclusion fails, choose a point of the lower child outside D . Every point of the upper child lies above it, hence is also outside the initial segment. The additional terminal factor does not change the ordering of the two children. $\square$

[merge] The next theorem is Corollary `cor:strict` of the ProveIt report *Every ordinal as a point-separating game value* (`ordinal_separation.tex`, lines 633–641), where it is proved through the point-separating game: an embedding $\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta$ with $\beta < \alpha$ would give a winning strategy of length β , contrary to that report’s exact game value. Source 02’s cylinder-fusion proof below is a second route, which avoids the game.

Theorem 5.3 (Strict hierarchy of binary cubes). *For all ordinals α, β ,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

Proof. Padding with zero bits proves the forward-length direction. Suppose instead that $f : \mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta$ and $\beta < \alpha$. Construct a source prefix s_ξ of length ξ , for $\xi \leq \beta$, so that f is constant on its first ξ target coordinates throughout C_{s_ξ} .

At a successor step, all previous target bits are fixed. The inverse image of target bit 0 at coordinate ξ , restricted to the current cylinder, is an initial segment. By lemma 5.2, one source child lies entirely on one side of this cut. Retain that child. At a limit stage take the union of the retained prefixes. Its full cylinder is the intersection of the earlier cylinders and remains nonempty. This is a fusion argument, not an appeal to finite-intersection compactness.

After β steps a source cylinder of length $\beta < \alpha$ still contains at least two points, while their images agree on all β coordinates. This contradicts injectivity. $\square$

Definition 5.4 (Binary lexicographic length). For a set-sized linear order L , define

$$\ell_2(L) = \min\{\theta : L \hookrightarrow \mathbb{B}_\theta\}.$$

This minimum exists in ZFC by lemma 5.1. A singleton and the empty order have length zero. Equimorphic orders have the same binary length.

Lemma 5.5 (Real and binary coordinates). $\mathbb{R} \simeq_e \mathbb{B}_\omega$. *More generally,*

$$\mathbb{R}_{\text{lex}}^\gamma \simeq_e \mathbb{B}_{\omega \cdot \gamma}$$

for every ordinal γ .

Proof. List the rationals as $(q_n)_{n < \omega}$ and send r to $j(r)(n) = 1$ exactly when $q_n < r$. This is an order embedding by the same coordinatewise argument as in lemma 5.1. In the other direction, send a bit sequence x to

$$T(x) = \sum_{n < \omega} \frac{2x(n)}{3^{n+1}}.$$

At the first unequal digit the contribution dominates the sum of all later possible differences, so this map is strictly increasing. Applying these maps coordinatewise and concatenating blocks of length ω proves the last statement. The ordinal length is $\omega \cdot \gamma$, not cardinal multiplication and not $\gamma \cdot \omega$. $\square$

For the uncountable initial ordinal $\kappa = 2^{\aleph_0}$,

$$\omega \cdot \kappa = \kappa. \quad (4)$$

Indeed, each $\omega \cdot \xi$ for $\xi < \kappa$ has cardinality less than κ , and their supremum is at least κ and at most κ . No regularity of κ is assumed.

[merge] The codes j and T of lemma 5.5 are also source 07's (07:436–458) and source 06's (06:433–445), and (4) is part of source 07's Lemma 3.2 (07:288), which also records $\kappa^\kappa = 2^\kappa$, $\sup_{\lambda < \kappa} \kappa^\lambda = 2^{<\kappa}$ and cf $\kappa > \omega$.

Corollary 5.6 (Strictness of real powers; Kuhlmann). *For all ordinals α, β ,*

$$\mathbb{R}_{\text{lex}}^\alpha \hookrightarrow \mathbb{R}_{\text{lex}}^\beta \iff \alpha \leq \beta.$$

First proof (from source 02's lemmas). By lemma 5.5, $\mathbb{R}_{\text{lex}}^\alpha \simeq_e \mathbb{B}_{\omega \cdot \alpha}$ and $\mathbb{R}_{\text{lex}}^\beta \simeq_e \mathbb{B}_{\omega \cdot \beta}$. By theorem 5.3, an embedding exists exactly when $\omega \cdot \alpha \leq \omega \cdot \beta$, that is, when $\alpha \leq \beta$, since left multiplication by ω is strictly increasing on ordinals. $\square$

Second route (source 07, Theorem 5.2, 07:477). Extension by zero gives the forward construction when $\alpha \leq \beta$. For the converse, suppose $\alpha > \beta$ and $F : \mathbb{R}_{\text{lex}}^\alpha \hookrightarrow \mathbb{R}_{\text{lex}}^\beta$. Recursively choose a source prefix s_δ of length δ and a target prefix t_δ of the same length, for $\delta \leq \beta$, such that every image of a source extension of s_δ extends t_δ .

At step $\delta < \beta$, subdivide the source cylinder according to its next real value r . Images of these subcylinders are ordered by r . Their δ -th target-coordinate sets are consequently ordered subsets of $\mathbb{R}$. At most countably many such sets contain two distinct values: for each such set choose a rational strictly between two of its values, and these rationals must be distinct. Choose an r outside the countable exceptional set. The coordinate is constant on this subcylinder, so it determines $t_{\delta+1}$.

At limits take unions. A source prefix of length at most β always has extensions, since $\beta < \alpha$. At the final step all extensions of s_β have exactly the same image, namely t_β . There are at least two such source extensions. This contradicts injectivity. $\square$

Source 07 identifies this ordinal-exponent theorem as classical: it is the ordinal case of Kuhlmann's embedding result [6, Corollary 2.4], and source 07's coordinate fusion is its own proof of the case needed. The point-separating report has only the binary case.

6 The exact ordinal spectrum

The upper bound is the essential point. Cardinality alone would permit well-ordered suborders far larger than κ .

Lemma 6.1 (Eventual constancy). *Let λ be an infinite cardinal. A nondecreasing sequence of length λ^+ with values in a linear order of cardinality at most λ is eventually constant. The same holds for a nonincreasing sequence.*

Proof. There are at most λ distinct values. Take the first occurrence of each value. Their supremum, increased by one if necessary, is below λ^+ : the supremum of at most λ ordinals below λ^+ has cardinality at most λ . After this bound no new value can first occur. Monotonicity prevents a change to a value that already occurred before the current value. Hence the sequence is constant on a tail. $\square$

[merge] Also source 06, Lemma 5.1 (06:454), for a regular λ and a target of cardinality below λ (“some fiber is unbounded, and a monotone unbounded fiber contains a tail”), and source 07, Lemma 4.3 (07:355).

Theorem 6.2 (Exhaustion forbids a successor-cardinal chain). *For every infinite linearly ordered set X of cardinality λ , $\mathcal{W}(X)$ contains neither λ^+ nor its reverse.*

Proof. Suppose $(e_\eta)_{\eta < \lambda^+}$ is strictly increasing. Recursively, for each $\xi < \lambda^+$, we shall obtain an injective prefix s_ξ of length ξ shared by every e_η on a tail of the index set.

The empty prefix starts the recursion. At a limit $\xi < \lambda^+$, the supremum of the preceding tail bounds is still below λ^+ , since $|\xi| \leq \lambda$. The union of the prefixes is injective and is shared by the resulting common tail.

Suppose the shared prefix s_ξ is exhaustive. Prefix-freeness forces all members of this tail to equal s_ξ , contrary to strict increase. Thus it is not exhaustive. Every enumeration on the tail has a value at coordinate ξ , and these values are nondecreasing because the earlier coordinates are fixed. By lemma 6.1, they are constant on a further tail. Their common value is not in $\text{ran } s_\xi$, by injectivity. Adjoin it to get $s_{\xi+1}$.

If the recursion never stops, its union is an injection $\lambda^+ \rightarrow X$, a contradiction. If it stops by exhaustion, we already have a contradiction. Replacing nondecreasing by nonincreasing proves the reverse-order assertion. $\square$

[merge] Also source 06, Theorem 5.2 (06:465), and source 07, Theorem 4.4 (07:367), by the same stabilization of a common prefix. Source 06 remarks (its Remark 5.3) that the regularity used at limit stages is a ZFC fact about the successor cardinal ν^+ , not an assumption about ν : a union of at most ν ordinals below ν^+ has cardinality at most $\nu \cdot \nu = \nu$; the continuum κ itself need not be regular. Source 07 notes that no separability of the alphabet is needed.

Remark 6.3. It would be incorrect to argue that the sequence of enumeration lengths is bounded below λ^+ . A long family can have unbounded lengths. The proof instead stabilizes a common prefix and uses the fact that an exhaustive prefix cannot be extended. This is exactly the extra step needed for the full order, rather than a fixed-length product.

Theorem 6.4 (Universal small orders and exact ordinals). *For every infinite linearly ordered set X of cardinality λ :*

- (a) *Every linear order of cardinality at most λ embeds into $\mathcal{W}_\lambda(X)$ and hence into $\mathcal{W}(X)$.*
- (b) *For every ordinal β ,*

$$\beta \hookrightarrow \mathcal{W}(X) \iff \beta < \lambda^+.$$

The same is true for reverse ordinals and for $\mathcal{W}_\lambda(X)$.

Proof. Compose cut coding with the pair-swap embedding in proposition 4.4. An ordinal has cardinality at most λ exactly when it is below λ^+ . Larger ordinals contain λ^+ as an initial segment and are excluded by theorem 6.2. Reverse orders are covered by the same small-order universality and the reverse half of that theorem. $\square$

For the real line, this proves the precise ordinal answer in theorem 3.1. Notice the distinction between the external lexicographic order on well-orderings and the well-order carried by any one of its elements. The external order contains long ascending and descending chains even though every element individually codes a well-order.

[merge] Also source 06, Corollary 5.4 (06:501), which states the spectrum for every slice $\mathcal{W}_\beta(X)$ of admissible length and notes that there is no largest embedded ordinal, and source 07, Corollary 4.5 (07:394). Source 07 adds (07:408–415) that $\kappa + 1$, $\kappa \cdot 2$ and $\kappa \cdot \kappa$ all embed while κ^+ does not, and that the map sending a well-order to its order type is onto $[\kappa, \kappa^+)$ but not monotone, so choosing one well-order of each type cannot give an increasing κ^+ -sequence. Source 06 adds (its Example 4.5) that, without CH, $\mathcal{W}$ already contains ω_1 , its reverse, every linear order on ω_1 , every ordinal of cardinality κ

and every continuum-sized ordered field or linear order, as orders, without preserving field operations; and (06:530–531) that $\mathcal{W}$ is not universal for orders of size 2^κ , since $\kappa^+ \leq 2^\kappa$.

7 Fixed length and the actual ultrafilter index

Theorem 7.1 (Fixed-length equivalences). *For every ultrafilter U on $\mathbb{N}$,*

$$\mathcal{W}_\kappa \simeq_e \mathbb{B}_\kappa \simeq_e \mathcal{A}_U \simeq_e \mathcal{A}.$$

Consequently, for any set-sized linear order L ,

$$L \hookrightarrow \mathcal{W}_\kappa \iff L \hookrightarrow \mathcal{A} \iff \ell_2(L) \leq \kappa. \quad (5)$$

Proof. The pair-swap construction gives $\mathbb{B}_\kappa \hookrightarrow \mathcal{W}_\kappa$. In the other direction, $\mathcal{W}_\kappa$ is a suborder of $\mathbb{R}_{\text{lex}}^\kappa$, which embeds into $\mathbb{B}_{\omega \cdot \kappa} = \mathbb{B}_\kappa$ by lemma 5.5.

Every ultrafilter U has cardinality κ . To see this, partition $\mathcal{P}(\mathbb{N})$ into complementary pairs $\{S, \mathbb{N} \setminus S\}$. There are κ such pairs, and an ultrafilter contains exactly one member of each pair. Choose a surjection $u : \kappa \rightarrow U$ and two distinct members $u_0 < u_1$ of U , in binary lexicographic order. For $x \in \{0, 1\}^\kappa$, set

$$a_x(2 \cdot \xi) = u_x(\xi), \quad a_x(2 \cdot \xi + 1) = u(\xi) \quad (\xi < \kappa). \quad (6)$$

Since $2 \cdot \kappa = \kappa$, these are maps of the required length. Their ranges are exactly U , and the first changed coding bit gives the first changed value. Thus $\mathbb{B}_\kappa \hookrightarrow \mathcal{A}_U$.

Finally, $\mathcal{A}_U \subseteq \mathcal{A} \subseteq (\mathcal{P}(\mathbb{N}))^\kappa$. Flattening the characteristic sequences of subsets of $\mathbb{N}$ turns the last order into $\mathbb{B}_{\omega \cdot \kappa} = \mathbb{B}_\kappa$. Composing these embeddings proves all the equivalences and (5). $\square$

The fact that even one fixed fiber $\mathcal{A}_U$ has this universality is an order-theoretic statement; it does not eliminate the role of all ultrafilters in the definable model construction. In particular, the embeddings in (6) use a chosen enumeration of U .

[merge] Also source 06, Proposition 4.6 (06:425), and source 07, (4) with its display (5.3) (07:456) and Theorem 11.2 (07:1129), where the ultrafilter coding is the same and the cardinality of an ultrafilter is its Lemma 11.1. Source 07 states the equimorphism with real coordinates as well: $\mathcal{W}_\kappa \simeq_e \mathcal{A} \simeq_e \mathbb{B}_\kappa \simeq_e \mathbb{R}_{\text{lex}}^\kappa$. Both stress that equimorphism is not isomorphism: $\mathbb{B}_\kappa$ has endpoints, whereas $\mathcal{W}_\kappa$ has none (06:447–448), and the characteristic-function order on $\mathcal{P}(\mathbb{N})$ has endpoints and adjacent pairs, unlike the usual order of $\mathbb{R}$, so a bijection between these alphabets is not an order isomorphism (source 07, Remark 1.2).

Corollary 7.2. *The binary lengths of $\mathcal{W}_\kappa$, $\mathcal{A}$, and $\mathcal{A}_U$ are all κ . Their embedded ordinals and reverse ordinals are exactly those below κ^+ . Moreover,*

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}_\kappa \iff \theta \leq \kappa, \quad \mathbb{R}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}_\kappa \iff \gamma \leq \kappa,$$

with the same criteria for $\mathcal{A}$ and $\mathcal{A}_U$.

Proof. Use theorems 5.3 and 7.1. For real powers, $\omega \cdot \gamma \leq \kappa$ is equivalent to $\gamma \leq \kappa$: (4) proves one direction, while $\gamma \geq \kappa + 1$ implies $\omega \cdot \gamma \geq \kappa + \omega > \kappa$. The ordinal spectrum follows from equimorphism with $\mathcal{W}_\kappa$ and theorem 6.4. $\square$

7.1 An exact finite-tail refinement

The last entry of an exhaustive enumeration is forced by all preceding entries. Several final entries, however, retain a genuine finite amount of lexicographic information. This gives a more precise hierarchy than the cardinality or ordinal spectrum alone detects.

For a positive integer m , write

$$D_{\kappa,m} = \mathbb{B}_\kappa \times_{\text{lex}} \{0, \dots, m-1\}.$$

The binary sequence is the first, more significant coordinate.

Theorem 7.3 (Finite-tail equivalence). *For every nonnegative integer n ,*

$$\mathcal{W}_{\kappa+n} \simeq_e D_{\kappa,n!}. \quad (7)$$

In particular,

$$\mathcal{W}_{\kappa+1} \simeq_e \mathcal{W}_\kappa, \quad \mathcal{W}_{\kappa+2} \simeq_e \mathbb{B}_{\kappa+1}.$$

Proof. Choose a fixed n -element set $F \subseteq \mathbb{R}$. Partition $\mathbb{R} \setminus F$ into κ ordered pairs, and use the pair-swap construction for the first κ entries. Append the r -th permutation of F in lexicographic order, for $0 \leq r < n!$. This embeds $D_{\kappa,n!}$ into $\mathcal{W}_{\kappa+n}$. The case $n = 0$ uses the unique empty permutation.

Conversely, code the first κ entries of an enumeration by a member of $\mathbb{B}_\kappa$, using rational-cut codes and (4). When two such prefixes agree, the remaining set of n reals is the same. Record the rank, between 0 and $n! - 1$, of the final permutation of those reals. The prefix code followed by this rank is an order embedding into $D_{\kappa,n!}$. $\square$

Theorem 7.4 (Finite-fiber capacity). *For every positive finite m ,*

$$\ell_2(D_{\kappa,m}) = \kappa + \lceil \log_2 m \rceil, \quad (8)$$

$$\mathbb{B}_\theta \hookrightarrow D_{\kappa,m} \iff \theta \leq \kappa + \lfloor \log_2 m \rfloor. \quad (9)$$

Hence the same formulas hold for $\mathcal{W}_{\kappa+n}$ with $m = n!$.

Proof. A finite increasing binary code for $\{0, \dots, m-1\}$ proves the upper bound in (8). Suppose $D_{\kappa,m} \hookrightarrow \mathbb{B}_{\kappa+j}$ for a finite j . Read the first κ target bits and apply lemma 5.2 successively to source cylinders $C_s \times \{0, \dots, m-1\}$. After κ steps, all m points over one source bit sequence have the same first κ target bits. They must fit into a fiber of size 2^j . Thus $m \leq 2^j$. Targets of length below κ are already excluded by their failure to contain $\mathbb{B}_\kappa$. This proves (8).

For (9), $2^j \leq m$ gives the direct embedding $\mathbb{B}_{\kappa+j} = \mathbb{B}_\kappa \times_{\text{lex}} \{0, 1\}^j \hookrightarrow D_{\kappa,m}$. Conversely, given an embedding in this direction, fuse source cylinders while reading the first κ bits of the target's $\mathbb{B}_\kappa$ coordinate. The final source cylinder contains 2^j points, but its target fiber has only m points. Hence $2^j \leq m$. Any ordinal length greater than $\kappa + \lfloor \log_2 m \rfloor$ contains the next forbidden finite extension as an initial coordinate block, so all larger lengths are also excluded. $\square$

Example 7.5. For $n = 3$ there are six final permutations. Thus

$$\ell_2(\mathcal{W}_{\kappa+3}) = \kappa + 3, \quad \mathbb{B}_\theta \hookrightarrow \mathcal{W}_{\kappa+3} \iff \theta \leq \kappa + 2.$$

The least coding length need not itself be a cube that embeds back. This small finite-fiber example foreshadows the much larger discrepancy for $\mathcal{W}$: its coding length is κ^+ , but $\mathbb{B}_{\kappa^+}$ does not embed.

8 Unbounded lengths and full lexicographic complexity

Lemma 8.1 (Arbitrarily long small-cardinal cubes). *Let X be an infinite linearly ordered set of cardinality λ . For every ordinal $\theta < \lambda^+$,*

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X).$$

Proof. Choose pairwise disjoint pairs $\{a_\xi, b_\xi\}$ for $\xi < \theta$, ordered so that $a_\xi < b_\xi$, leaving a residual set of cardinality λ . This is possible because $|\theta| \leq \lambda$ and λ is infinite. For each bit sequence, list each pair in its selected orientation, in the ordinal order of θ , and then append one fixed enumeration of the residual set. The total length is $2 \cdot \theta + \lambda$, an ordinal of cardinality λ . First changed bits give first changed entries. $\square$

[merge] Also the second half of source 06's Lemma 4.1 (06:351) and, with real coordinates, source 07's Proposition 5.1 (07:460).

Theorem 8.2 (The full coding length). *For every infinite linearly ordered set X of cardinality λ ,*

$$\ell_2(\mathcal{W}(X)) = \lambda^+, \quad \mathbb{B}_\theta \hookrightarrow \mathcal{W}(X) \iff \theta < \lambda^+. \quad (10)$$

In particular, $\mathcal{W}$ embeds into $\mathbb{B}_{\kappa^+}$, but the reverse embedding does not exist.

Proof. Embed X into $\mathbb{B}_\lambda$ by lemma 5.1. For an enumeration of length $\alpha < \lambda^+$, concatenate its coordinate codes into a binary sequence of length $\lambda \cdot \alpha$, and pad with zeros to length λ^+ . Since $|\lambda \cdot \alpha| \leq \lambda$, this is a legitimate shorter initial block. Prefix-freeness ensures that two different exhaustive enumerations disagree before either ends, so padding introduces no ambiguity. This proves the upper bound $\ell_2(\mathcal{W}(X)) \leq \lambda^+$.

If $\mathcal{W}(X)$ embedded into $\mathbb{B}_\delta$ for $\delta < \lambda^+$, then lemma 8.1 would give $\mathbb{B}_{\delta+1} \hookrightarrow \mathbb{B}_\delta$, contrary to theorem 5.3. This proves equality.

The positive cube embeddings are lemma 8.1. To exclude $\mathbb{B}_{\lambda^+}$, note that this cube contains a copy of λ^+ : map $\eta < \lambda^+$ to the sequence with value 1 below η and 0 from η onward. This would contradict theorem 6.2. Larger cubes contain this forbidden cube. $\square$

[merge] Source 07 proves the real-coordinate counterpart, $\text{repr}_\mathbb{R}(\mathcal{W}) = \kappa^+$ and $\text{repr}_\mathbb{R}(\mathcal{W}_\kappa) = \text{repr}_\mathbb{R}(\mathcal{A}) = \kappa$ (its Theorem 5.3, 07:505), from corollary 5.6 by the same padding argument; the two invariants are compared in table 2.

For the real case, the upper coding construction can use countable rational-cut blocks rather than blocks of length κ . Thus an enumeration of length α is first coded at length $\omega \cdot \alpha$. The ultimate common padding length is still κ^+ .

Corollary 8.3 (No bounded-length universal suborder). *Fix $\tau < \kappa^+$. Then*

$$\mathcal{W} \not\hookrightarrow \bigcup_{\alpha \leq \tau} \mathcal{W}_\alpha.$$

Consequently, the image of every order embedding $\mathcal{W} \hookrightarrow \mathcal{W}$ has an unbounded set of enumeration lengths in κ^+ .

Proof. The bounded union embeds into $\mathbb{B}_{\omega \cdot (\tau+1)}$ by coding and padding. Its target length is below κ^+ , whereas $\ell_2(\mathcal{W}) = \kappa^+$. $\square$

Corollary 8.4 (No fixed power below κ^+ ; sources 02 and 07). *$\mathcal{W}$ embeds neither into $\mathcal{W}_\kappa$, nor into $\mathcal{A}$, nor into any real power $\mathbb{R}_{\text{lex}}^\gamma$ or binary cube $\mathbb{B}_\gamma$ with $\gamma < \kappa^+$. This holds in ZFC, without any hypothesis on $2^{<\kappa}$.*

First proof (source 02). $\mathcal{W}_\kappa$ is the stratum $\alpha = \kappa$, so corollary 8.3 with $\tau = \kappa$ excludes it; $\mathcal{A} \simeq_e \mathcal{W}_\kappa$ by theorem 7.1. An embedding into $\mathbb{R}_{\text{lex}}^\gamma$ would give $\mathcal{W} \hookrightarrow \mathbb{B}_{\omega \cdot \gamma}$ by lemma 5.5, with $\omega \cdot \gamma < \kappa^+$, contrary to $\ell_2(\mathcal{W}) = \kappa^+$ (theorem 8.2). $\square$

Second route (source 07, Corollary 5.4, 07:525). By real representation rank: $\text{repr}_\mathbb{R}(\mathcal{W}) = \kappa^+$, whereas $\text{repr}_\mathbb{R}(\mathcal{W}_\kappa) = \text{repr}_\mathbb{R}(\mathcal{A}) = \kappa$ and $\text{repr}_\mathbb{R}(\mathbb{R}_{\text{lex}}^\gamma) = \gamma$ by corollary 5.6. $\square$

[merge] This answers, unconditionally, source 06’s Research question 14.1 (06:1332), which asked whether $\mathcal{W}$ embeds into $\mathcal{W}_\kappa$ or into one fixed power $\mathbb{R}_{\text{lex}}^\gamma$ with $\gamma < \kappa^+$ “in any model of ZFC”. Source 06 had the negative answer for $\mathcal{W}_\kappa$ only under $2^{<\kappa} < 2^\kappa$; its argument is kept as a different route in the remark below. Its question about the longer fixed slices $\mathcal{W}_\beta$ is answered as well: $\mathcal{W}$ embeds into no bounded union of strata (corollary 8.3).

Remark 8.5 (The cellularity route; source 06, Corollary 9.5). If $2^{<\kappa} < 2^\kappa$, the separation $\mathcal{W} \not\hookrightarrow \mathcal{W}_\kappa$ also follows from cellularity. The full order has 2^κ pairwise disjoint six-element convex blocks (proposition 12.1). An order embedding into $\mathcal{W}_\kappa$ would send, from each block, three selected points to an ordered triple. The intervals between the first and third images would be nonempty and pairwise disjoint. This contradicts $\text{cell}(\mathcal{W}_\kappa) = 2^{<\kappa}$ (proposition 12.2). The proof uses intervals between images, not continuity of the embedding. When $2^{<\kappa} = 2^\kappa$ this cardinal-invariant argument says nothing; source 06 left that case open, and corollary 8.4 closes it.

Corollary 8.6 (Real-coordinate powers). *For every ordinal γ ,*

$$\mathbb{R}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W} \iff \gamma < \kappa^+.$$

Proof. For $\gamma < \kappa^+$, the ordinal $\omega \cdot \gamma$ also lies below κ^+ , so use lemmas 5.5 and 8.1. For $\gamma \geq \kappa^+$, restricting all coordinates to two reals gives a copy of $\mathbb{B}_{\kappa^+}$ and therefore of κ^+ . $\square$

[merge] Also source 07, Proposition 5.1 and the last assertion of Corollary 5.4 (07:460, 07:525), and, for the positive part, source 06, Proposition 4.6 (06:425).

These results give an exact classification for binary and real-coordinate lexicographic powers. They do not classify every linear order of cardinality 2^κ that has no κ^+ -chain in either direction. In particular, absence of the two long monotone chains is not asserted to be a sufficient embedding criterion for $\mathcal{W}$.

9 An exact classification of every fixed ordinal type

The finite-tail result extends to *every* possible ordinal length, not only to $\kappa + n$. The key is to exploit the shrinking alphabet after a long prefix has already enumerated most of the ground set. Coding every remaining real by a fresh block of ω bits would miss this improvement.

Lemma 9.1 (Countable alphabets). *If X is a countable linear order, then X_{lex}^ω embeds into $\mathbb{B}_\omega$. Consequently, for every nonzero limit ordinal λ , X_{lex}^λ embeds into $\mathbb{B}_\lambda$.*

Proof. There are countably many nonempty finite words t over X . For a word of length m , the set

$$D_t = \{x \in X^\omega : x \upharpoonright m \leq_{\text{lex}} t\}$$

is an initial segment. Enumerate these sets and code a point by bit 0 when it belongs to the corresponding D_t , and bit 1 otherwise. If $x < y$ first differ at coordinate $m - 1$, then $t = x \upharpoonright m$ gives a cut containing x but not y . Every cut code is monotone, so the resulting binary code is an order embedding.

Every limit ordinal has the form $\lambda = \omega \cdot \eta$. Apply the countable-alphabet code separately to its successive ω -blocks. The concatenated binary length is $\omega \cdot \eta = \lambda$. $\square$

Theorem 9.2 (All fixed-type strata). *Let $X \subseteq \mathbb{R}$ be infinite, with the inherited order, and let $|\alpha| = |X|$. Write*

$$\alpha = \lambda + n, \quad \lambda \text{ a nonzero limit ordinal, } n < \omega.$$

Then

$$\mathcal{W}_\alpha(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} \{0, \dots, n! - 1\}. \quad (11)$$

In particular, this is an exact equimorphism classification of all strata $\mathcal{W}_\alpha$ of the real well-ordering space.

Proof. The lower embedding has a uniform construction. Fix an n -element set $F \subseteq X$ and partition $X \setminus F$ into pairs indexed by λ . Orient these pairs according to a member of $\mathbb{B}_\lambda$ and append one of the $n!$ permutations of F . Since λ is a limit ordinal, $2 \cdot \lambda = \lambda$, so the result has exactly the required length $\lambda + n$.

For the upper embedding, induct on the infinite cardinal $\mu = |X|$, proving the assertion simultaneously for all inherited real suborders of that size and all appropriate α . If $\mu = \aleph_0$, code the first λ entries by [lemma 9.1](#), and rank the permutation of the remaining n elements. This gives the required target.

Suppose μ is uncountable and the assertion is known for smaller infinite cardinals. Use ordinal division to write

$$\alpha = \mu \cdot \eta + \rho, \quad 0 \leq \rho < \mu, \quad \eta > 0,$$

and put $\delta = \mu \cdot \eta$. Since $\omega \cdot \mu = \mu$, $\omega \cdot \delta = \delta$. Rational-cut coding therefore embeds the first δ real entries into $\mathbb{B}_\delta$ without increasing their ordinal length.

For a fixed such prefix p , the residual set $Y = X \setminus \text{ran } p$ has cardinality $|\rho|$. If ρ is finite, its tail is just one of $\rho!$ permutations, which finishes the construction. If ρ is infinite, write $\rho = \sigma + n$, with σ a limit ordinal. By the induction hypothesis, choose an embedding

$$\mathcal{W}_\rho(Y) \hookrightarrow \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, n! - 1\}.$$

Choose such an embedding for each possible residual set Y ; this is a set-sized use of Choice. Append its binary part after the prefix code and retain its final finite rank. The target is

$$\mathbb{B}_{\delta+\sigma} \times_{\text{lex}} \{0, \dots, n! - 1\} = \mathbb{B}_\lambda \times_{\text{lex}} \{0, \dots, n! - 1\}.$$

Two different prefixes are compared inside the first δ bits. Equal prefixes have the same residual set and are compared by the chosen embedding of their tails. Thus the assembled map preserves and reflects order. The induction is complete. $\square$

Corollary 9.3 (Exact length and cube spectrum in every stratum). *For $\alpha = \lambda + n$ as above,*

$$\begin{aligned} \ell_2(\mathcal{W}_\alpha(X)) &= \lambda + \lceil \log_2(n!) \rceil, \\ \mathbb{B}_\theta \hookrightarrow \mathcal{W}_\alpha(X) &\iff \theta \leq \lambda + \lfloor \log_2(n!) \rfloor. \end{aligned}$$

For $X = \mathbb{R}$, every stratum still has cardinality 2^κ and ordinal spectrum precisely $\{\beta : \beta < \kappa^+\}$.

Proof. The finite-fiber fusion proof of [theorem 7.4](#) works with any ordinal λ in place of κ ; it only uses full binary cylinders through λ steps. Apply it to [theorem 9.2](#). For the real case, $|\lambda| = \kappa$ and $\lambda \geq \kappa$, so the stratum contains $\mathbb{B}_\kappa$ and is a suborder of $\mathcal{W}$. The cardinality and ordinal conclusions follow from the earlier upper and lower bounds. $\square$

Corollary 9.4 (The exact comparison between strata). *Let $\alpha = \lambda + n$ and $\beta = \mu + m$ have cardinality κ , with λ, μ nonzero limit ordinals and $m, n < \omega$. Then*

$$\mathcal{W}_\alpha \hookrightarrow \mathcal{W}_\beta \iff \lambda < \mu \text{ or } [\lambda = \mu \text{ and } n! \leq m!].$$

Thus the fixed strata are linearly ordered by embeddability. Distinct strata are equimorphic exactly for the pairs $\mathcal{W}_\lambda, \mathcal{W}_{\lambda+1}$ with λ a limit ordinal.

Proof. For the same limit length, fusing source cylinders while reading the λ target bits forces an entire source fiber of size $n!$ into a target fiber of size $m!$. Hence $n! \leq m!$ is necessary, and it is plainly sufficient. If $\lambda < \mu$, then $\lambda + \lceil \log_2(n!) \rceil < \mu$, since μ is a limit ordinal. Code the source into that shorter cube and then into $\mathbb{B}_\mu$ in the target. If $\mu < \lambda$, the target codes into a cube of length $\mu + \lceil \log_2(m!) \rceil < \lambda$, whereas the source contains $\mathbb{B}_\lambda$. The strict cube hierarchy excludes an embedding. $\square$

Example 9.5. For the following infinite-tail types, the fixed stratum is equimorphic with the binary cube of exactly the same length:

$$\mathcal{W}_{\kappa+\omega} \simeq_e \mathbb{B}_{\kappa+\omega}, \quad \mathcal{W}_{\kappa+\omega^2} \simeq_e \mathbb{B}_{\kappa+\omega^2}, \quad \mathcal{W}_{\kappa \cdot 2} \simeq_e \mathbb{B}_{\kappa \cdot 2}.$$

More generally, $\mathcal{W}_\lambda \simeq_e \mathcal{W}_{\lambda+1} \simeq_e \mathbb{B}_\lambda$ for every limit λ of cardinality κ . The formula is about embeddability, not an order isomorphism with a cube that has endpoints.

Proposition 9.6 (Density and cellularity across the strata). *For every $\kappa \leq \alpha < \kappa^+$,*

$$d(\mathcal{W}_\alpha) = \text{cell}(\mathcal{W}_\alpha) = \begin{cases} 2^{<\kappa}, & \alpha = \kappa, \kappa + 1, \kappa + 2, \\ 2^\kappa, & \alpha \geq \kappa + 3. \end{cases}$$

Proof. The first three cases are proved in [proposition 12.4](#). For the others, write $\alpha = \kappa + \tau$, where $\tau \geq 3$. Choose a set $Y \subseteq \mathbb{R}$ of cardinality $|\tau|$ whose complement still has cardinality κ . There are 2^κ distinct exhaustive prefixes of length κ listing $\mathbb{R} \setminus Y$. Each corresponding cylinder in $\mathcal{W}_\alpha$ is isomorphic to $\mathcal{W}_\tau(Y)$ and has at least three points. Inside each choose a nonempty open interval between three such points. These intervals are pairwise disjoint. Cardinality gives the upper bound. $\square$

10 Self-absorption and closure under lexicographic constructions

The availability of arbitrary exhaustive lengths lets one concatenate entire well-orderings, rather than squeezing them into a fixed coordinate budget.

Lemma 10.1 (Many disjoint real copies). *There are κ pairwise disjoint subsets of $(0, 1)$, each admitting an order isomorphism from $\mathbb{R}$ onto that subset.*

Proof. Use the increasing injection $j : \mathbb{R} \rightarrow \{0, 1\}^\omega$ from [lemma 5.5](#). For $t \in \{0, 1\}^\omega$, interleave the fixed bits of t with the variable bits of $j(r)$:

$$z_{t,r}(2n) = t(n), \quad z_{t,r}(2n+1) = j(r)(n).$$

For fixed t , first disagreements are those of $j(r)$, so $r \mapsto T(z_{t,r})$ is increasing. Different t give disjoint images because ternary coding is injective. An affine rescaling puts all images strictly inside $(0, 1)$. There are $2^{\aleph_0} = \kappa$ parameters t . $\square$

Theorem 10.2 (Sharp power absorption). *For every ordinal γ ,*

$$\mathcal{W}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W} \iff \gamma < \kappa^+. \quad (12)$$

For $0 < \gamma < \kappa^+$, $\mathcal{W}_{\text{lex}}^\gamma \simeq_e \mathcal{W}$.

Proof. Assume $0 < \gamma < \kappa^+$. Choose disjoint images $X_\xi = h_\xi[\mathbb{R}] \subseteq (0, 1)$ for $\xi < \gamma$, with each h_ξ increasing. Let

$$Z = \mathbb{R} \setminus \bigcup_{\xi < \gamma} X_\xi,$$

and choose a fixed exhaustive enumeration z of Z . The set Z has cardinality κ , because it contains $\mathbb{R} \setminus (0, 1)$. For a tuple $(e_\xi)_{\xi < \gamma} \in \mathcal{W}^\gamma$, define

$$E((e_\xi)_{\xi < \gamma}) = \bigwedge_{\xi < \gamma} (h_\xi \circ e_\xi) \frown z. \quad (13)$$

All images being disjoint, this is an injective enumeration of all reals. Its ordinal length is the sum of at most κ ordinals, each of cardinality at most κ , followed by the length of z . The total has cardinality κ , so it is an allowed member of $\mathcal{W}$.

For two distinct tuples, let ξ be their first different component. Earlier components are equal as exhaustive enumerations, including their lengths; hence the concatenated prefixes are identical. Within component ξ , prefix-freeness supplies a disagreement before either component ends. Since h_ξ is increasing, that disagreement has exactly the required sign. Thus E is an order embedding.

The empty product is a singleton and embeds trivially. For $\gamma \geq \kappa^+$, choose two points of $\mathcal{W}$ and use them as binary coordinate values. The product then contains $\mathbb{B}_{\kappa^+}$ and hence κ^+ , which $\mathcal{W}$ cannot contain. Finally, for $\gamma > 0$, varying one coordinate and fixing all other coordinates embeds $\mathcal{W}$ into the product. $\square$

[merge] Also source 07, Lemma 6.1 and Theorem 6.2 (07:548, 07:563), by the same interleaved copies of $\mathbb{R}$ and the same concatenation. Source 07 stresses that the copies must be disjoint *ordered* copies of $\mathbb{R}$, not disjoint intervals, of which only countably many exist (07:543–546), and that the ranges need not be convex.

Corollary 10.3 (Closure of the embedding class). *Let $\mathcal{E}(\mathcal{W}) = \{L : L \hookrightarrow \mathcal{W}\}$. This class is closed under reversal, under ordinal-indexed lexicographic products of at most κ factors, and under lexicographic sums whose index order and every summand belong to $\mathcal{E}(\mathcal{W})$.*

Proof. Negation of all real entries reverses $\mathcal{W}$, so the class is closed under reversal. Embed each factor into $\mathcal{W}$ and apply [theorem 10.2](#) for products. For a sum $\sum_{i \in I} L_i$, choose embeddings $f : I \rightarrow \mathcal{W}$ and $g_i : L_i \rightarrow \mathcal{W}$. Then

$$(i, x) \mapsto (f(i), g_i(x))$$

embeds the sum into $\mathcal{W} \times_{\text{lex}} \mathcal{W}$, and the latter embeds into $\mathcal{W}$. The family of chosen embeddings is a set-sized choice in ZFC. $\square$

[merge] Also source 07, Corollary 6.3 (07:599).

Corollary 10.4 (Failure of absorption at fixed length). *For every ordinal γ ,*

$$(\mathcal{W}_\kappa)_{\text{lex}}^\gamma \simeq_e \mathbb{B}_{\kappa \cdot \gamma}, \quad \mathcal{A}_{\text{lex}}^\gamma \simeq_e \mathbb{B}_{\kappa \cdot \gamma}.$$

Either product embeds back into its original factor exactly when $\gamma \leq 1$.

Proof. Equimorphisms can be applied coordinatewise, and consecutive blocks of κ binary coordinates have ordinal length $\kappa \cdot \gamma$. For $\gamma \geq 2$, this length exceeds κ ; use [theorem 5.3](#). The cases 0 and 1 are immediate. $\square$

[merge] Source 07 states the same equimorphism with real coordinates, $(\mathcal{W}_\kappa)_{\text{lex}}^\beta \simeq_e \mathbb{R}_{\text{lex}}^{\kappa \cdot \beta}$ and $\mathcal{A}_{\text{lex}}^\beta \simeq_e \mathbb{R}_{\text{lex}}^{\kappa \cdot \beta}$ for $\beta > 0$ (its Proposition 6.4 and Theorem 11.2, 07:613, 07:1129). This is the same statement: $\mathbb{R}_{\text{lex}}^{\kappa \cdot \beta} \simeq_e \mathbb{B}_{\omega \cdot \kappa \cdot \beta}$ and $\omega \cdot \kappa \cdot \beta = \kappa \cdot \beta$ by (4). Source 07 adds the rank $\text{repr}_{\mathbb{R}}((\mathcal{W}_\kappa)_{\text{lex}}^\beta) = \text{repr}_{\mathbb{R}}(\mathcal{A}_{\text{lex}}^\beta) = \kappa \cdot \beta$.

10.1 Many convex copies inside the full order

The preceding embeddings need not have convex images. There is also a particularly strong convex version for a large family of copies.

Proposition 10.5. *$\mathcal{W}$ contains 2^κ pairwise disjoint open convex suborders, each isomorphic to $\mathcal{W}$. They can be chosen so that their index order is $\mathbb{B}_\kappa$.*

Proof. Let $I = (0, 1)$. Partition $\mathbb{R} \setminus I$ into κ ordered pairs. For $x \in \{0, 1\}^\kappa$, let s_x be the pair-swap enumeration of $\mathbb{R} \setminus I$. The cylinders C_{s_x} are pairwise disjoint and ordered according to x . By [lemma 4.3](#), each is isomorphic to $\mathcal{W}(I)$, and an increasing bijection $\mathbb{R} \rightarrow I$ identifies $\mathcal{W}(I)$ with $\mathcal{W}$. The order $\mathcal{W}(I)$ has no endpoints, since I has none. Thus every point of a cylinder is bracketed by two points in that cylinder, making the convex cylinder open in $\mathcal{W}$. $\square$

11 Adjacent pairs and finite condensation

Lemma 11.1 (Extrema of exhaustive enumeration orders). *For any nonempty linearly ordered set Y , $\mathcal{W}(Y)$ has a least element exactly when the given order on Y is a well-order. That element is the increasing enumeration of Y . It has a greatest element exactly when the given order on Y is a reverse well-order, in which case it is the decreasing enumeration.*

Proof. In a lexicographically least exhaustive enumeration, the value at each coordinate must be the least as-yet-unused element. Otherwise a smaller unused choice, followed by any exhaustive enumeration of the residual set, would give a smaller member of $\mathcal{W}(Y)$. Thus a least enumeration must be increasing and exhibit Y as a well-order. Conversely, at the first changed coordinate of any other enumeration, the increasing enumeration uses the least available value, so it is lexicographically smaller. Reversing the ground order proves the maximum assertion. $\square$

[merge] Also source 06, Lemma 3.3 (06:263), and the first half of source 07, Lemma 8.1 (07:714).

Theorem 11.2 (Complete adjacency criterion). *Let $e < f$ in $\mathcal{W}(X)$, where X is any infinite linearly ordered set. Let s be their maximal common prefix, of length δ , and write*

$$Y = X \setminus \text{ran } s, \quad a = e(\delta), \quad b = f(\delta).$$

Then e and f are adjacent if and only if all of the following hold:

- (a) Y is finite;
- (b) $a < b$ are consecutive elements of Y in its given order;
- (c) after a , the enumeration e lists $Y \setminus \{a\}$ in decreasing order;
- (d) after b , the enumeration f lists $Y \setminus \{b\}$ in increasing order.

In particular, an adjacent pair has equal enumeration lengths and differs only in a finite final segment.

Proof. If $a < c < b$ for some $c \in Y$, then an exhaustive extension of $s \frown c$ lies strictly between e and f . If e is not the greatest member of its a -cylinder, a larger member of that cylinder also lies between them. Likewise, f must be the least member of its b -cylinder. By lemma 11.1, $Y \setminus \{a\}$ must be reverse well-ordered and $Y \setminus \{b\}$ must be well-ordered.

The intersection $Y \setminus \{a, b\}$ is consequently both well-ordered and reverse well-ordered. It is finite: an infinite well-order contains an increasing ω -sequence, whose range has no greatest member and cannot be reverse well-ordered. Thus Y is finite. The two extremal tails are then exactly the decreasing and increasing finite lists in the statement.

Conversely, any point between e and f would have to extend s , by convexity. Its next entry could not lie strictly between a and b . Choosing a cannot exceed the greatest tail e , and choosing b cannot precede the least tail f . No intermediate point exists. $\square$

[merge] Also source 06, Theorem 6.2 (06:548), and source 07, Theorem 8.2 (07:747), by the same argument. Source 07 states the criterion without condition (a) and derives finiteness of Y ; its conditions (ii) and (iii) are (c) and (d) here, through lemma 11.1.

Example 11.3 (A jump and an isolated point). Choose any exhaustive prefix s enumerating $\mathbb{R} \setminus \{a, b\}$, with $a < b$. Then

$$s \frown (a, b) < s \frown (b, a)$$

is an adjacent pair in the full order $\mathcal{W}$, not just in a fixed-length suborder.

If $a < b < c$ and s enumerates $\mathbb{R} \setminus \{a, b, c\}$, its cylinder is the six-point chain

$$\begin{aligned} s \smallfrown (a, b, c) &< s \smallfrown (a, c, b) < s \smallfrown (b, a, c) \\ &< s \smallfrown (b, c, a) < s \smallfrown (c, a, b) < s \smallfrown (c, b, a). \end{aligned}$$

The four interior points are isolated in the order topology of $\mathcal{W}$.

[merge] This is also source 06's Example 6.1 and its Appendix B. Source 06 reads the middle jump $s \smallfrown (b, c, a) < s \smallfrown (c, a, b)$ through [theorem 11.2](#): the old suffix after b is decreasing (c, a) , the new suffix after c is increasing (a, b) , and b, c are adjacent in the unused alphabet. It warns that the example must not be extrapolated to all points: an enumeration of length κ has no finite final tail, lies in a singleton finite-condensation class, and has uncountable character on both sides; nor should a permutation generator be read as an algorithm that enumerates all well-orderings of $\mathbb{R}$.

Definition 11.4. The finite-condensation relation on a linear order is

$$x \sim_{\text{fin}} y \iff [\min(x, y), \max(x, y)] \text{ is finite.}$$

It is a convex equivalence relation.

Theorem 11.5 (Exact finite blocks). *Let $e \in \mathcal{W}$, and write its length uniquely as $\alpha = \lambda + n$, where λ is a nonzero limit ordinal and $n < \omega$. If $n = 0$, its finite-condensation class is a singleton. If $n > 0$, its class consists exactly of all $n!$ permutations of the final n reals after the fixed prefix $e \upharpoonright \lambda$.*

These classes are the maximal finite convex blocks. A point is isolated exactly when it is an interior point of one of the blocks with $n \geq 3$.

Proof. The displayed permutations form a finite convex cylinder. If a point is adjacent to e , [theorem 11.2](#) says that the first disagreement leaves only finitely many entries. Hence the two enumerations have the same length and agree before its final finite remainder n . No adjacency can cross to a different displayed block. Every finite interval is connected by finitely many adjacency steps, so the displayed block is the full finite-condensation class.

In a linear order with no endpoints, a point is isolated exactly when it has both an immediate predecessor and an immediate successor. The first and last points of these maximal blocks have no adjacent point beyond the block; an interior point has both required neighbors. A factorial block has an interior point precisely when $n! \geq 3$, equivalently $n \geq 3$. $\square$

[merge] Also source 06, Proposition 6.3 and Corollary 6.4 (06:590, 06:607), and source 07, Theorem 8.3 (07:770). Source 06 counts the isolated points of a block: exactly its $n! - 2$ nonextremal permutations. Source 07 makes explicit why an adjacency chain cannot leave the block: for the length $\lambda + n$, a common prefix with finitely many unused values has length at least λ , since adding finitely many positions to an ordinal below the limit λ cannot reach λ .

The quotient by finite condensation is a dense linear order: if two different classes were adjacent in the quotient, their union would be a finite interval and they would be the same class. We do not need to assume that this quotient is isomorphic to any familiar generalized real line. Its precise isomorphism type remains a separate question.

[merge] Sources 06 and 07 prove that $\mathcal{W}/\sim_{\text{fin}}$ is equimorphic with $\mathcal{W}$ (source 06, Theorem 11.1, 06:1133; source 07, Proposition 10.5, 07:1072), answering the question asked in source 02's [section 16](#). Its isomorphism type is still open.

12 Order topology, size of intervals, and cofinalities

12.1 Global invariants of the full order

Proposition 12.1. *The order $\mathcal{W}$ has no endpoints, is isomorphic to its reverse, and satisfies*

$$\text{cf}(\mathcal{W}) = \text{ci}(\mathcal{W}) = \omega, \quad \text{d}(\mathcal{W}) = \text{cell}(\mathcal{W}) = 2^\kappa.$$

It has exactly 2^κ isolated points.

Proof. Every real can be the first entry of an exhaustive enumeration. Choosing first entries tending to $+\infty$ gives a countable cofinal sequence; choosing first entries tending to $-\infty$ gives a countable coinital sequence. Neither endpoint exists. The map $e \mapsto -e$, with the same ordinal domain, is an order-reversing bijection.

Fix three reals and use all pair-swap enumerations of their complement as prefixes of length κ . For each prefix choose one of the four interior permutations in its six-point block. This gives 2^κ isolated points by [theorem 11.5](#). Each isolated singleton must be met by a dense set, and the singletons are pairwise disjoint nonempty open sets. The upper bounds follow from $|\mathcal{W}| = 2^\kappa$. $\square$

[merge] Also source 06, Corollary 3.6 and Theorem 6.6 (06:320, 06:650), and source 07, Lemma 8.1 and Corollary 8.4 (07:714, 07:798). Both add the weight: $w(\mathcal{W}) = 2^\kappa$, since the interval base has at most $|\mathcal{W}|^2 = 2^\kappa$ members and $\text{cell} \leq \text{d} \leq w$. Source 06 also shows that the isolated points are dense; sources 06 and 07 obtain $\text{cf } \mathcal{W} = \text{ci } \mathcal{W} = \omega$ from the stronger fact that every residual order $\mathcal{W}(Y)$, $Y \subseteq \mathbb{R}$, has cofinality and coinitality in $\{1, \omega\}$.

There is a second, independent reason for the large cellularity: [proposition 10.5](#) gives 2^κ disjoint open copies of $\mathcal{W}$. Thus the phenomenon is not solely a by-product of isolated finite tails. For example, selecting one point from each finite-condensation class still leaves 2^κ disjoint nontrivial interval regions coming from those large cylinders.

It would nevertheless be false to say that *every* interval in $\mathcal{W}$ is large. Some intervals are empty, some are finite, and a cylinder leaving countably infinitely many unused reals has cardinality only $2^{\aleph_0}$. The residual-set lemma, not the ambient cardinality, controls such examples.

12.2 The dense fixed-length order

Proposition 12.2. *The order $\mathcal{W}_\kappa$ is dense, has no endpoints, and is self-dual. Its global cofinality and coinitality are ω , and*

$$\text{d}(\mathcal{W}_\kappa) = \text{cell}(\mathcal{W}_\kappa) = 2^{<\kappa}. \quad (14)$$

At every point e , the cofinality of the left side and the coinitality of the right side are both $\text{cf}(\kappa)$.

Proof. Let $e < f$ first disagree at $\delta < \kappa$. The common prefix has used fewer than κ reals, whereas the interval $(e(\delta), f(\delta))$ has cardinality κ . Choose an unused real strictly between the two entries. Extending the resulting prefix to a bijection of length κ gives an intermediate point. This proves density. Endpoints, global cofinalities, and self-duality are treated as for $\mathcal{W}$.

There are at most

$$\sum_{\xi < \kappa} \kappa^{|\xi|} = 2^{<\kappa}$$

prefixes of length below κ . Here finite nonzero exponents contribute $\kappa = 2^{\aleph_0}$, and for infinite $\mu < \kappa$, $\kappa^\mu = (2^{\aleph_0})^\mu = 2^\mu$. Choose one exhaustive extension for each such prefix. The resulting set is dense, since the intermediate prefix in the previous paragraph determines a whole cylinder inside the given interval.

For every infinite cardinal $\mu < \kappa$, orient μ disjoint pairs to obtain 2^μ incompatible prefixes of length $2 \cdot \mu < \kappa$. Their cylinders are nonempty, convex, and have no endpoints: one can change a later coordinate upward or downward while fewer than κ values have been used. The cylinders are therefore open. This proves the lower cellularity bound after taking the supremum over $\mu < \kappa$. Cellularity never exceeds density, proving (14).

Finally, for every prefix length $\xi < \kappa$, there are points on both sides of e agreeing with it below ξ : change a still later coordinate to an unused smaller or larger real and complete the enumeration. Choose such points at a cofinal set of $\text{cf}(\kappa)$ prefix lengths. They are cofinal on the left and coinital on the right. Conversely, fewer than $\text{cf}(\kappa)$ points on one side have a bounded set of first-disagreement coordinates with e . A perturbation agreeing longer than that bound lies closer to e than all of them. This proves both local lower bounds. $\square$

[merge] Also source 06, Theorems 9.2 and 9.3 (06:973, 06:1001), and source 07, Theorem 7.1 (07:637), by the same short-prefix arguments. Both add $w(\mathcal{W}_\kappa) = 2^{<\kappa}$, since the short-prefix cylinders, at most $2^{<\kappa}$ of them, are clopen and form a base.

Remark 12.3 (Regularity is not assumed). The formula uses $\text{cf}(\kappa)$, not automatically κ . However, $\text{cf}(2^{\aleph_0}) > \omega$. Here is a direct diagonal proof. If $\{0, 1\}^\omega = \bigcup_{n < \omega} X_n$ with each $|X_n| < 2^{\aleph_0}$, partition ω into infinite sets I_n . Choose a pattern on I_n that is not the restriction of any member of X_n . The union of these patterns belongs to none of the X_n , a contradiction. A countable cofinal sequence of smaller cardinals would produce just such a cover. Consequently $\mathcal{W}_\kappa$ is not first countable at any point.

[merge] Sources 06 (06:1029–1038) and 07 (Lemma 3.2, 07:305–310) prove $\text{cf } \kappa > \omega$ through König's inequality: if cardinals $\kappa_n < \kappa$ were cofinal in κ , then $\kappa = \sum_n \kappa_n < \prod_n \kappa_n^+ \leq \kappa^{\aleph_0} = \kappa$, a contradiction; the strict inequality is the same diagonal argument.

12.3 A topology refinement for finite tails

Proposition 12.4. *For finite $n \geq 0$, the following hold.*

- (a) *For $n = 0, 1$, $\mathcal{W}_{\kappa+n}$ is dense and has no isolated points.*
- (b) *For $n = 2$, it has 2^κ adjacent pairs and no isolated points. Its density and cellularity are still $2^{<\kappa}$.*
- (c) *For $n \geq 3$, it has 2^κ isolated points and hence density and cellularity 2^κ .*

For $n = 0, 1$, density and cellularity are $2^{<\kappa}$ as well. The weight of $\mathcal{W}_{\kappa+2}$ is 2^κ , even when its density is smaller.

Proof. If two points first disagree below κ , the unused-real argument in [proposition 12.2](#) gives an intermediate point of the same length. If their first κ entries agree, they lie in one finite permutation block. The block sizes are $1, 1, 2, 6, \dots$, proving the assertions about adjacency and isolation. Varying a pair-swap prefix while fixing the final finite set gives 2^κ blocks.

For $n \leq 2$, a nonempty open interval cannot be finite: a finite open set in a T_1 space would contain an isolated point. Such an interval therefore contains two points with different prefixes below κ , since each full κ -prefix has at most two extensions. As before, a shorter cylinder fits between those points. Choosing one extension of every prefix of length below κ gives the upper density bound, and incompatible short prefixes give the lower cellularity bound.

For the weight assertion, every adjacent pair $a < b$ requires a basic open set containing b and contained in (a, ∞) . Different adjacent pairs require different such basic sets: if $a < b \leq c < d$, one set cannot both contain b and be contained in (c, ∞) . Thus any base has size at least 2^κ . The set of all endpoint intervals supplies the reverse bound. $\square$

13 Finer properties of the Kanovei–Shelah index

Equimorphism with $\mathbb{B}_\kappa$ gives the exact embeddable orders, but does not determine global cofinality or topology. Those require separate arguments.

Lemma 13.1. *Every nonempty cylinder in $\mathcal{A}$ determined by fewer than κ coordinates has no endpoints. In particular, $\mathcal{A}$ is a dense linear order without endpoints.*

Proof. Fix $a \in \mathcal{A}_U$ and a prefix of length $\xi < \kappa$. Every ultrafilter contains $\mathbb{N}$ and has κ members. Since the prefix lists fewer than κ values, arbitrarily far out in the tail there is a value other than $\mathbb{N}$. Replacing such a value by $\mathbb{N}$, and then enumerating all of U on the remaining coordinates, gives a larger point with the prescribed prefix.

For a smaller point, if U has a least member, the tail cannot consist only of this member; choose a later larger value and replace it by the least one. If U has no least member, replace any chosen later value by a smaller member of U . In either case complete the remaining coordinates by an enumeration of U . Thus there are both larger and smaller points in the cylinder.

If $a < b$ first differ at δ , apply the preceding construction to a with its prefix through δ fixed. A larger extension in that cylinder still lies below b . $\square$

Proposition 13.2. *The full index $\mathcal{A}$ satisfies*

$$|\mathcal{A}| = 2^\kappa, \quad d(\mathcal{A}) = \text{cell}(\mathcal{A}) = 2^{<\kappa}, \quad \text{cf}(\mathcal{A}) = \text{cf}(\kappa), \quad \text{ci}(\mathcal{A}) = \omega.$$

At every point, the left and right local cofinalities are both $\text{cf}(\kappa)$. In particular, $\mathcal{A}$ is not isomorphic to its reverse and is not isomorphic to $\mathcal{W}_\kappa$.

Proof. Cardinality follows from the embeddings in [theorem 7.1](#). For density, there are at most $2^{<\kappa}$ prefixes of length below κ . By [lemma 13.1](#), every nonempty interval contains a cylinder with such a prefix; choose one extension for each extendible prefix. Conversely, fix U and two of its members. All binary choices for a prefix of an infinite length $\mu < \kappa$ extend to members of $\mathcal{A}_U$. Their full $\mathcal{A}$ -cylinders are disjoint nonempty open sets. This proves the cellularity and density formulas.

Let $\nu(a)$ be the first coordinate at which a is not $\mathbb{N}$. It exists because $\text{ran } a$ is an ultrafilter with κ members. For a cofinal set of ordinals $\delta < \kappa$, construct a map beginning with δ copies of $\mathbb{N}$, then a fixed non- $\mathbb{N}$ member of a fixed ultrafilter, then an exhaustive tail. These maps are cofinal in $\mathcal{A}$. A family of fewer than $\text{cf}(\kappa)$ maps has its values of $\nu(a)$ bounded below κ ; a sufficiently long initial string of $\mathbb{N}$ lies above the whole family. Hence $\text{cf}(\mathcal{A}) = \text{cf}(\kappa)$.

For each natural number n , take a member of $\mathcal{A}$ whose first value is $\{n\}$ and whose range is the principal ultrafilter at n . The singletons $\{n\}$ descend coininitially among the nonempty subsets of $\mathbb{N}$ in binary lexicographic order. Therefore these maps are coininitial in $\mathcal{A}$, which has no least element. This proves $\text{ci}(\mathcal{A}) = \omega$.

Local cofinalities follow from the same bounded-disagreement argument used for $\mathcal{W}_\kappa$, now using [lemma 13.1](#) for perturbations after arbitrary prescribed prefixes. Finally, $\text{cf}(\kappa) > \omega$, so the two global cofinalities are unequal. They rule out self-duality and also rule out an isomorphism with $\mathcal{W}_\kappa$. $\square$

[merge] Also source 07, Theorem 11.3 (07:1163), with the same cofinality and coinitality witnesses; source 07 adds the weight, a clopen base of short prefixes, and coinitality witnesses for the free-ultrafilter variant of $\mathcal{A}$.

For comparison, a fixed principal-ultrafilter fiber $\mathcal{A}_U$ has coinitality $\text{cf}(\kappa)$: it has a least available coordinate value, which can be repeated for arbitrarily long initial segments. A nonprincipal fiber has coinitality ω , because the cofinite sets form a decreasing coinital sequence in its coordinate order. All fibers nevertheless have the same embeddability class by [theorem 7.1](#).

14 Explicit countable gaps and Dedekind completion

14.1 A gap caused by a repeated real

Proposition 14.1. *Both $\mathcal{W}$ and $\mathcal{W}_\kappa$ have an unfilled cut of cofinality type (ω, ω) .*

Proof. Divide either order into the following initial and final segments:

$$\begin{aligned} L &= \{e : e(0) < 0 \text{ or } [e(0) = 0 \text{ and } e(1) < 0]\}, \\ R &= \{e : e(0) > 0 \text{ or } [e(0) = 0 \text{ and } e(1) > 0]\}. \end{aligned}$$

Every exhaustive enumeration has at least two entries, and injectivity excludes $e(0) = e(1) = 0$. Thus this is a partition and $L < R$. For $n \geq 1$, complete $(0, -1/n)$ and $(0, 1/n)$ to exhaustive enumerations of length κ . The first family is cofinal in L ; the second is coinital in R . Neither side has an endpoint, and a point between all the bounds would have to start $(0, 0)$, which is forbidden. $\square$

[merge] Also source 07, Example 7.3 (07:703), with the same bounds; source 07 concludes that neither order has the countable interpolation property of an η_1 -order.

This also illustrates why large local character at every point does not imply countable saturation as a dense order. The obstruction is an omitted cut, rather than a point with small local character.

14.2 A gap caused by incompatible ultrafilter values

Proposition 14.2. *The order $\mathcal{A}$ also has an unfilled (ω, ω) cut.*

Proof. Let u be the even natural numbers and v the odd natural numbers. Put

$$v_n^- = (v \setminus \{2n + 1\}) \cup \{2n + 2\}, \quad v_n^+ = v \cup \{2n\}.$$

In binary lexicographic order, v_n^- increases toward v from below, and v_n^+ decreases toward v from above. Each pair u, v_n^- belongs to the principal ultrafilter at $2n + 2$; each pair u, v_n^+ belongs to the principal ultrafilter at $2n$. Therefore both two-term prefixes (u, v_n^-) and (u, v_n^+) extend to members of $\mathcal{A}$.

A map between all these bounds must have first value u and second value v . Indeed, if a binary sequence differs from v , its first differing bit is eventually shared by neither bounding side in the required direction. But no ultrafilter contains the disjoint sets u and v . Thus the cut is not realized. The initial segment of maps with first value below u , or first value u and second value below v , has the lower sequence cofinal; the complementary final segment has the upper sequence coinital. $\square$

[merge] Source 07 proves the same statement with different bounds, and adds a gap of character (cf κ , cf κ) (its Proposition 11.4, 07:1218).

These statements concern the index orders. They do not contradict the countable saturation of the elementary extension constructed from the index in the Kanovei–Shelah theorem.

14.3 Completion does not create longer ordinals

We use the Dedekind completion that fills unrealized two-sided cuts and keeps existing points and jumps. Adding endpoints is optional and does not change the following cardinal or ordinal conclusions.

Lemma 14.3 (Cuts have small cofinal witnesses). *Every nonempty initial segment of $\mathcal{W}$ has a cofinal subset of cardinality at most κ . Every nonempty final segment has a coinital subset of cardinality at most κ .*

Proof. If an initial segment had no such cofinal subset, at every stage $\xi < \kappa^+$ one could choose a point larger than all previously chosen points: the preceding set has cardinality at most κ and is not cofinal. This constructs the forbidden increasing sequence of length κ^+ . Reverse the order for the other assertion. $\square$

[merge] Also source 06, Corollary 5.5 (06:520), for every nonempty suborder of $\mathcal{W}(X)$, and source 07, Corollary 4.6 (07:417).

Theorem 14.4 (Completion invariants). *Let $\widehat{\mathcal{W}}$ be a Dedekind completion of $\mathcal{W}$. Then*

$$|\widehat{\mathcal{W}}| = 2^\kappa, \quad \beta \hookrightarrow \widehat{\mathcal{W}} \iff \beta < \kappa^+, \quad \ell_2(\widehat{\mathcal{W}}) = \kappa^+.$$

The ordinal equivalence also holds for reverse ordinals. Every two-sided cut of $\mathcal{W}$ has each of its two cofinalities at most κ .

Proof. By lemma 14.3, every lower half of a cut is determined by a cofinal subset of size at most κ . Thus the number of cuts is at most

$$|\mathcal{W}|^\kappa = (2^\kappa)^\kappa = 2^\kappa.$$

The lower cardinal bound follows from the copy of $\mathcal{W}$ already present.

Suppose the completion contained an increasing sequence of length κ^+ . Between the first and third members of each successive three-point block, choose a point of $\mathcal{W}$. Such a point always exists: if the middle point is old, use it; if it is an added cut, the cut property supplies an old point between the outer endpoints. The resulting points form a strictly increasing sequence of length κ^+ in $\mathcal{W}$, a contradiction. The same argument applies in reverse. The positive ordinal embeddings were already present in $\mathcal{W}$.

For binary length, $\mathcal{W} \hookrightarrow \widehat{\mathcal{W}}$ gives the lower bound. The cube $\mathbb{B}_{\kappa^+}$ is Dedekind complete, as verified below. An embedding $f : \mathcal{W} \rightarrow \mathbb{B}_{\kappa^+}$ extends over each unrealized cut (L, R) by assigning it $\sup f[L]$. This value is strictly above every $f(l)$ because L has no greatest element, and strictly below every $f(r)$ because R has no least element. Distinct cuts remain distinct, by an old point between them. Thus the extension is an order embedding. If endpoints are to be added, first put $f[\mathcal{W}]$ inside a proper binary cylinder by a fixed finite prefix and then use the ambient endpoints. Its length remains κ^+ because finite $+\kappa^+ = \kappa^+$. $\square$

[merge] The cardinality also follows from source 07's count of initial segments (Corollary 4.6, 07:417) and from source 06's count of proper gaps (Theorem 7.6, 06:824), which shows that there are exactly 2^κ of them. Source 06 also proves the ordinal assertion by the same interval argument (06:851–856).

Lemma 14.5 (Completeness of binary lexicographic cubes). *Every $\mathbb{B}_\theta$ is Dedekind complete and has both endpoints.*

Proof. The constant sequences give endpoints. For a nonempty subset S , construct its supremum by retaining a nonempty set of candidates with a common prefix. At a successor step choose bit 1 if some current candidate has bit 1, and retain those candidates; otherwise choose bit 0 and retain all current candidates. At a limit, intersect the candidate sets.

If the first empty intersection occurs at a limit $\lambda \leq \theta$, complete the accumulated prefix by zeros. Every member of S was discarded at an earlier first disagreement where it had 0 and the chosen prefix had 1, so the completed sequence is an upper bound. Any strictly smaller sequence first disagrees with this upper bound before λ ; at that stage there was a candidate with the chosen 1, which is larger than the proposed smaller bound. Thus the completed sequence is the least upper bound. If no empty intersection occurs through θ , the final sequence belongs to S and is its maximum. This covers all cases. $\square$

Remark 14.6. The completion theorem does not prove $\widehat{\mathcal{W}} \hookrightarrow \mathcal{W}$. Both orders have the same cardinality, ordinal spectrum, and binary length, but these invariants do not in general decide embeddability. Whether the completion can be absorbed is one of the natural next questions.

15 Foundational scope and interpretation

15.1 A definable order is not a selected well-order

The set of all well-order relations on $\mathbb{R}$ is definable from $\mathbb{R}$, and first-difference comparison can be expressed without choosing one member of that set as privileged. This observation does not produce a definable well-ordering of $\mathbb{R}$. It defines a linear order *on the collection* of all its well-orderings. Likewise, forming all the presentations in (2) does not select one ultrafilter or one presentation.

Our proofs nevertheless use Choice. It supplies well-orderings of the real line, partitions into pairs, selected enumerations of residual sets, and simultaneous choices of embeddings. In ZF alone, $\mathbb{R}$ need not be well-orderable, so the literal collection $\mathcal{W}$ may be empty. Even when that collection is nonempty, the arguments using families of choices should not automatically be read as proofs in a weaker theory.

The set-sized nature of the construction is also essential. Every possible length is below κ^+ , and every well-ordering is a subset of $\mathbb{R}^2$. There is no proper class of elements hidden in $\mathcal{W}$, despite its unbounded range of lengths below κ^+ . In contrast, the phrase “all ordinals” denotes a proper class and does not describe a suborder of this set.

15.2 Which aspects are genuinely independent?

The proofs separate four kinds of information. The ordinal spectrum records how far a single monotone well-ordered chain can run. Binary length measures the depth needed to distinguish all points through ordered binary cuts. Adjacency records what happens when only finitely many ground elements remain. Finally, topology and cut cofinalities describe local approach and missing limits. These quantities interact, but none should be silently substituted for another.

For example, $\mathcal{W}_\kappa$ and $\mathcal{A}$ have identical embedding classes yet unequal global cofinalities. The full order and its completion have the same ordinal spectrum and binary length, but differ in completeness. The strata $\mathcal{W}_{\kappa+2}$ and $\mathcal{W}_{\kappa+3}$ have different binary lengths and different finite blocks, though their cardinalities and ordinal spectra coincide. Under some cardinal-arithmetic assumptions even density fails to distinguish those last two orders; their isolated points still do.

All embeddings in this article are purely order-theoretic. An order embedding need not be continuous for the two order topologies. In particular, the fact that a cube or a complete order embeds into a dense order does not transfer its order-topological compactness to the ambient space. No topological or model-theoretic consequence is inferred from an embedding without a separate argument.

16 Further research questions and topics

The following are questions proposed by this investigation. We have not verified that every one is open in the published literature. The solved fixed-stratum *equimorphism* problem is deliberately excluded from the list; the next issues concern sharper structure and broader universality.

16.1 An intrinsic embedding criterion for the full order

Find a necessary and sufficient condition on a set-sized linear order L for $L \hookrightarrow \mathcal{W}$. We have an exact answer when $\ell_2(L) \leq \kappa$ and exact answers for the binary and real-coordinate powers, but not for every order of cardinality 2^κ and binary length κ^+ . Absence of κ^+ and its reverse is necessary; this article does not prove it sufficient.

A concrete route is to study trees of convex partitions whose branches consume distinct symbols from an alphabet of size κ . The exhaustion argument shows what a forbidden branch would do. The remaining

problem is to characterize when a compatible coding can be organized across all branches, not merely along each branch separately.

16.2 Can the full order absorb its completion?

Decide whether

$$\widehat{\mathcal{W}} \hookrightarrow \mathcal{W}.$$

The affirmative direction would make the two orders equimorphic, since the other embedding is automatic. The invariants in [theorem 14.4](#) do not obstruct it, and the power-absorption theorem suggests considerable room for coding cuts. A proof must nevertheless assign codes to different cuts coherently. Independently choosing a cofinal sequence for each cut need not preserve their lexicographic comparison.

A related problem asks whether the dense finite-condensation quotient $\mathcal{W}/\sim_{\text{fin}}$ is equimorphic with $\mathcal{W}$, and whether its completion has a more recognizable presentation.

16.3 Isomorphism types beyond equimorphism

Determine the order-isomorphism types of individual strata and their natural quotients. In particular, equimorphism of $\mathcal{W}_\lambda$ and $\mathcal{W}_{\lambda+1}$ does not settle whether they are isomorphic. Their presentations impose different exhaustion conditions, and an isomorphism need not preserve the represented ordinal length.

The comparison theorem [corollary 9.4](#) supplies a complete baseline: the task is no longer to decide which fixed strata embed, but to identify additional invariants capable of distinguishing equimorphic ones. Cut spectra, automorphism orbits, and local characters are plausible candidates.

16.4 Automorphisms and recovery of the prefix structure

Every increasing bijection $h : \mathbb{R} \rightarrow \mathbb{R}$ induces an automorphism $e \mapsto h \circ e$ of $\mathcal{W}$. Determine the full automorphism group and whether some canonical fragment of the prefix tree can be reconstructed from the external linear order alone.

Finite-condensation classes and their sizes are intrinsic and therefore preserved. It is less clear how much of the non-finite ordinal remainder of an enumeration can be recovered. One useful intermediate target is a classification of the orbits of endpoints and interior points of the finite blocks, together with the singleton-block points.

16.5 Exact spectra of cuts and point characters

Classify all pairs of left and right cofinalities occurring at points and unrealized cuts of $\mathcal{W}$ and its fixed strata. We have computed the local pair (cf κ , cf κ) throughout $\mathcal{W}_\kappa$, exhibited missing (ω, ω) cuts, and bounded every side of a full-order cut by κ . Those facts leave many cases undecided.

A productive decomposition may separate three sources of cuts: a missing next coordinate in an ordered residual set, a limit of increasingly long injective prefixes, and exhaustion at the end of a finite block. This could also lead to exact criteria for convex suborders to be complete, separable, or first countable.

16.6 Other ground orders

The ordinal cutoff and full binary length were proved for every infinite linearly ordered ground set X . The all-strata classification used $X \subseteq \mathbb{R}$, particularly countable separating cuts and rational coordinate codes. Determine the weakest hypotheses on X under which

$$\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$$

still holds. For orders with large intrinsic binary length, develop a replacement formula that records both coordinate complexity and the shrinking size of the residual alphabet.

This question naturally includes nonseparable real-closed fields and set-sized suborders of surreal numbers. The distinction between the underlying order and any algebraic structure must remain explicit.

16.7 Ultrafilter fibers and the hyperreal presentation

Among nonprincipal ultrafilters U , determine whether the order-isomorphism type of $\mathcal{A}_U$ depends on U , even though its embedding class does not. Principal and nonprincipal fibers are already distinguished by their coinitalities, so an undifferentiated claim that all fibers are isomorphic would be false.

One can also ask which transformations of the index preserve a specified finite-support ultrapower presentation, rather than merely providing an order embedding of indices. The index order by itself does not encode all ultrafilter data used by the construction; equimorphism of index orders is not a proof of isomorphism of the associated nonstandard models.

16.8 Weak choice and uniform definability

Determine the choice principles needed for the separate results: existence of a nonempty exhaustive order, the successor-cardinal chain obstruction, universality for small orders, optimal fixed-stratum coding, and power absorption. Distinguish existence of arbitrary embeddings from embeddings definable uniformly from their input orders.

Herzberg, Kanovei, Katz, and Lyubetsky develop a modification of the hyperreal construction with transfer under countable choice and a further well-ordering hypothesis for properness [3]. Their work provides motivation for separating the choice strength of the index-order analysis from that of the nonstandard model. No equivalence with their precise axiomatic hypotheses is asserted here.

16.9 Cardinal arithmetic and stability of the classification

The embeddability theorems are stated in ZFC, but their numerical interpretation depends on the continuum. Investigate which finer isomorphism invariants depend on $2^{<\kappa}$, 2^κ , or $\text{cf}(\kappa)$. For instance, the density distinction between the first three real strata and later ones is strict exactly when $2^{<\kappa} < 2^\kappa$; the adjacency and isolation distinctions do not require that inequality.

A model-comparison study should specify whether the real set, the collection of its well-orderings, or only selected cardinal parameters are preserved. Keeping the reals fixed does not by itself keep every subset of $\mathbb{R}^2$, and hence every member of $\mathcal{W}$, fixed.

16.10 A proof-assistant development for ProveIt

A formalization can be divided into reusable modules. First formalize prefix-free ordinal words and their first-difference order. Then establish cut coding, the full-cylinder obstruction, and the successor-cardinal stabilization argument. On that foundation, formalize ordinal block concatenation and finite-fiber capacity, followed by the cardinal induction for all fixed strata. Adjacency and finite condensation form a largely independent final module.

The most delicate obligations are not numerical. They are the existence of common tail bounds at limit stages, correct ordinal rather than cardinal multiplication, and preservation of a full cylinder under transfinite fusion. The finite verification program accompanying this article checks none of these infinitary obligations and should not be presented as a substitute for them.

A Proof dependencies and reproducible finite checks

The central arguments have a short dependency structure:

$$\begin{aligned} \text{prefix-freeness} + \text{eventual constancy} &\implies \text{exact ordinal cutoff,} \\ \text{full-child lemma} + \text{fusion} &\implies \text{strict cube hierarchy,} \\ \text{pair swaps} + \text{cut coding} &\implies \text{small-order universality,} \\ \text{countable codes} + \text{cardinal induction} &\implies \text{all fixed-stratum equivalences,} \\ \text{disjoint real copies} + \text{exhaustive concatenation} &\implies \text{full power absorption.} \end{aligned}$$

These arguments are self-contained apart from elementary ZFC facts about well-ordering, cardinal arithmetic, and transfinite recursion, which are explained at their points of use.

The archive contains `finite_checks.py`, a Python program using only the standard library, together with its actual output in `finite_checks.json`. Run it with

```
python finite_checks.py -output finite_checks.json
```

It checks pair-swap order preservation, cut coding of finite labeled orders, every candidate adjacent pair of permutations through size six, consecutive permutation pairs through size eight, finite concatenation, and the integer capacities used in the terminal-factor formulas. The supplied run passes **552,450** exact checks. Counts and parameter bounds are recorded by the program, rather than inferred from an informal estimate.

[merge] In this report the program is shipped as `code/02-strata-finite_checks.py` and its recorded output as `data/02-strata-finite_checks.json`; the text of the program still names its delivered output `finite_checks.json`. Its default output path is `finite_checks.json` in the current directory, so a rerun should pass an explicit scratch path, for example `pycode/02-strata-finite_checks.py --output<scratch>/finite_checks.json`, and never the shipped `data/` file. A rerun on a copy during placement reproduced the recorded JSON, identical after removing the carriage returns that Python writes in text mode on Windows. The finite checks of sources 06 and 07 are shipped as `code/06-spectra-finite_checks.py` and `code/07-rank-finite_checks.py`, with their recorded outputs in `data/`; they too test finite instances only.

What the checks establish. They are exact finite tests of the stated constructions and permutation formulas. They do not verify a transfinite theorem, establish a literature-priority claim, or constitute a Lean or Rocq formalization. The mathematical evidence for the infinite claims is the written proof, subject to ordinary review.

B Source audit and manuscript status

The Kanovei–Shelah definition was checked directly in the paper, including the displayed index and its comparison rule on page 2 of the arXiv PDF. [merge] Source 07 locates the same definition in [1, §2, p. 160], on the printed page of the published version (its audit, item 1), whereas source 02 cites Section 1 of the paper; both quote the same index and the same comparison rule. The section numbering was not rechecked at the merge. The paper’s bibliographic information was cross-checked against Shelah’s official paper archive. The weak-choice follow-up was checked through its authors’ arXiv record. Knoblauch’s work is cited for the related research terminology and bibliographic context, not as an uninspected source for any specific theorem proved here.

The ProveIt source was inspected at the immutable commit

6ea60e3677bd847a00baf023c3c532c83884530e.

Its relevant report is the binary-cube and point-separation manuscript cited below. Only its elementary cylinder perspective is reused; the required lemma is reproved. No existing formal verification of the new article’s theorems is claimed, and this work does not modify the repository.

The literature search was targeted rather than exhaustive. It identified related primary sources, but it does not settle historical priority for the classification and absorption statements. [merge] Sources 06 and 07 record their own audits in the shipped files `06-spectra-research_audit.md` and `07-rank-source_audit.md`, both pinned to the same ProveIt revision; neither claims priority, and both describe their repository search as targeted. The questions in [section 16](#) are proposed next steps, not an authenticated list of published open problems. All websites and repository material were consulted for this manuscript on 2 October 2026.

C Theorem crosswalk of the three sources

[merge] Every numbered statement of the three sources, by topic. Source numbers are those of the delivered PDFs (T theorem, L lemma, P proposition, C corollary, D definition, E example, R remark, Q research question), followed by the line of the delivered `.tex`. “Here” is where the statement is printed in this report; “n.y.p.” marks a result of sources 06 or 07 that is not yet printed at this stage of the merge.

Table 4: [merge] Crosswalk.

Result	Source 02	Source 06	Source 07	Here
Main results at a glance	T2.1 (191)	T1.1 (102)	T2.1 (187)	3.1
Definition; prefix freeness; linear order	L3.1 (269)	L2.1 (171), D2.2, P2.3 (198)	D1.1 (118), §1.2	4.1
Cylinders and residual sets; next-value sum	D3.2, L3.3 (297)	D3.1, L3.2 (246)	L3.1 (259)	4.3 , (3)
Cardinality 2^λ by pair swaps	P3.4 (318)	L4.1 (331), C4.2 (363)	L4.1 (329), C4.6 (417)	4.4
Cut (initial-segment) coding	L4.1 (337)	L4.3 (373)	L4.2 (339)	5.1
Full child; strict cube hierarchy	L4.2, T4.3 (367)	—	—	5.3 (= PSG cor:strict)
Binary coding length ℓ_2	D4.4 (393)	—	—	5.4
Real and binary codes	L4.5 (402)	in P4.6	(5.1)–(5.3)	5.5
Cardinal identities; cf $\kappa > \omega$	02:425, R11.3	in T9.3	L3.2 (288)	(4), 12.2
Strict real powers (Kuhlmann)	—	—	T5.2 (477)	5.6
Eventual constancy	L5.1 (437)	L5.1 (454)	L4.3 (355)	6.1
No ν^+ -chain	T5.2 (451), R5.3	T5.2 (465), R5.3	T4.4 (367)	6.2
Universality; ordinal spectrum	T5.4 (487)	C5.4 (501), E4.5	L4.2, C4.5 (394)	6.4
Universality in every slice $\mathcal{W}_\beta(X)$	—	T4.4 (389)	—	n.y.p.
Separating families of initial segments	—	06:415–423	—	n.y.p.
$\mathcal{W}_\kappa \simeq_e \mathbb{B}_\kappa \simeq_e \mathcal{A}$	T6.1 (516), C6.2	P4.6 (425)	(5.3), L11.1, T11.2 (1129)	7.1
Finite tails $\mathcal{W}_{\kappa+n}$; capacity	T6.3, T6.4, E6.5	—	—	7.3 , 7.4
Small cubes and real powers below κ^+	L7.1 (665), C7.4 (732)	L4.1, P4.6	P5.1 (460)	8.1 , 8.6

Result	Source 02	Source 06	Source 07	Here
Coding length $\ell_2(\mathcal{W}) = \kappa^+$	T7.2 (682)	—	—	8.2
Representation ranks $\text{repr}_{\mathbb{R}}$ $\mathcal{W}$ into no fixed power below κ^+	— C7.3 (719)	— C9.5 (1053), Q14.1	T5.3 (505) C5.4 (525)	n.y.p. 8.3, 8.4, 8.5
All fixed strata	L8.1, T8.2 (782), C8.3–C8.4, E8.5, P8.6	—	—	9.2–9.6
Disjoint copies; power absorption; closure	L9.1, T9.2 (946), C9.3	—	L6.1, T6.2 (563), C6.3	10.2, 10.3
Fixed-length powers	C9.4 (1005)	—	P6.4 (613)	10.4
$\mathcal{W}_{\text{lex}}^{\omega} \simeq_e \mathcal{W}$ but not $\cong$	—	—	07:628–633	n.y.p.
Convex copies of $\mathcal{W}$	P9.5 (1026)	—	—	10.5
Extremal enumerations	L10.1 (1044)	L3.3 (263)	L8.1 (714)	11.1
Countable monotone real sequences; countable ends	—	L3.4, L3.5 (294), C3.6	L8.1 (714)	n.y.p.
Adjacency criterion; example	T10.2 (1062), E10.3	E6.1, T6.2 (548), App. B	T8.2 (747)	11.2
Finite blocks; isolated points	D10.4, T10.5 (1129)	P6.3 (590), C6.4	T8.3 (770)	11.5
Dense isolated points; perfect remainder	—	L6.5, T6.6 (650), P6.7	C8.4 (798)	n.y.p.
Global invariants of $\mathcal{W}$	P11.1 (1166)	C3.6, T6.6	C8.4	12.1
Topology of $\mathcal{W}_{\kappa}$	P11.2 (1204), R11.3	L9.1, T9.2, T9.3, R9.4	T7.1 (637), C7.2	12.2; extras n.y.p.
Finite-tail strata topology	P11.4 (1264)	—	—	12.4
Topology of $\mathcal{A}$	L12.1, P12.2 (1330)	—	T11.3 (1163)	13.2; extras n.y.p.
(ω, ω) gaps of $\mathcal{W}, \mathcal{W}_{\kappa}, \mathcal{A}$	P13.1 (1384), P13.2	—	E7.3 (703), P11.4	14.1, 14.2
Small cuts; completion; complete cubes	L13.3, T13.4, L13.5, R13.6	C5.5, T7.6 (824)	C4.6	14.3–14.5
Gap spectrum of $\mathcal{W}$	topic (1635)	D7.1, L7.2, T7.3, L7.4, T7.5 (802)	L10.1, T10.2 (941)	n.y.p.
Point-character spectrum	topic (1635)	T8.1, P8.2, T8.3 (913)	P9.1, T9.2 (861)	n.y.p.
Gaps of $\mathcal{W}_{\kappa}$; no convex copy	—	P10.3 (1101)	T10.3 (996), C10.4	n.y.p.
Lexicographic-sum decomposition	—	D10.1, T10.2 (1078)	—	n.y.p.
Dense quotient $\mathcal{W}/\sim_{\text{fin}}$	question (1604)	T11.1 (1133)	P10.5 (1072)	n.y.p.
Bounded surreal sign orders	—	T12.1 (1198)	—	n.y.p.
Further gaps and fibres of $\mathcal{A}$	—	—	P11.4 (1218)	n.y.p.
Well-ordering-based ultrafilter index	—	—	T12.1 (1265)	n.y.p.
Research questions	10 topics (1573)	Q14.1–Q14.7	Q14.1–Q14.7	16
Summary tables	Table 1	App. A	§2 table	3

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